

Trace-Tempered Choice

A Self-Contained Axiomatisation with Explicit Payoffs and Records

Daniele Condorelli
University of Warwick

Draft of 9th August 2026

Abstract

An action can matter twice: through the payoff it produces and through the visible trace it leaves. We separate these roles in the primitive domain. A finite channel maps actions into pairs consisting of a payoff-bearing outcome and an explicit record. Continuous weak order, coherent comparison environments, data processing on both sides of the channel, and a branchwise sure-thing principle characterize

$$\mathbb{E}[u(O)] + \lambda I(A; O, R), \quad \lambda > 0.$$

No primitive payoff–trace separability axiom is required. It is derived from the temporal asymmetry of the domain: records may arrive first, while payoff is terminal. The proof isolates that derivation in a calibrated branch-insertion and finite-telescope argument. Appendix A proves the mutual-information characterization on the pure-trace restriction as the first part of the single proof of the theorem. Every primitive axiom is shown to be independent of the others.

Contents

1	Introduction	3
2	Related Literature	5
3	The Model	7
4	The Representation Theorem	10
5	Choice implications	13
5.1	The trace test	13
5.2	Deliberate randomisation	14
5.3	The price of being on the record	16
5.4	What preferences cannot reveal	18
6	Interpretation and scope	19
	Proofs	24
A	Pure trace on the constant-payoff domain	24
A.1	Proof	25
A.1.1	Stage 1: posterior-law quotient and affine fibre values	25
A.1.2	Stage 2: transport across priors and support faces	29

A.1.3	Stage 3: products and a common comparison scale	30
A.1.4	Stage 4: sequential decomposition and coherent face scales	33
A.1.5	Stage 5: compatibility and entropy identification	36
B	Completion of the representation theorem	41
B.1	Preliminaries	41
B.2	The two common scales	43
B.3	The branch calculation	44
B.4	Proof of the theorem	46
C	Auxiliary results	48
C.1	Sign, uniqueness, and matching	48
C.2	Independence of the axioms	49
C.3	Proofs of the choice implications	50

1 Introduction

Shower upon him every earthly blessing . . . He would even risk his cakes and would deliberately desire the most fatal rubbish, the most uneconomical absurdity . . . simply in order to prove to himself—as though that were so necessary—that men still are men and not the keys of a piano. . . . If you say that all this, too, can be calculated and tabulated . . . then man would purposely go mad in order to be rid of reason and gain his point!

— Fyodor Dostoevsky, *Notes from the Underground*, Part I, Chapter VIII [22]

Dostoevsky’s provocation has the structure of a choice problem. The cakes represent material payoff, and the Underground Man is willing to risk them so that his conduct can attest to something: that it is his own rather than the mechanical implication of a table. Yet the proof he seeks is self-defeating. An act fixed in advance cannot provide it, and deliberate perversity offers no escape: rebellion, too, can be “calculated and tabulated.” His final resort is therefore paradoxical: he would “purposely go mad in order to be rid of reason”—not because madness is what he values, but because it is his last imagined way of preventing the table from closing before he acts. This paper takes that evidentiary idea literally, in a deliberately narrow statistical sense. In Dostoevsky’s example, conduct is observed directly: the act supplies its own evidence. We consider the more general case in which an act is observed only through a possibly noisy payoff-bearing outcome and an explicit visible record.

For visible consequences to carry information about an act, two conditions are needed. There must be uncertainty beforehand about which act will occur, and the eventual payoff–record pair must discriminate statistically among the possible acts. Either condition can fail independently. A lottery over acts leaves no trace when every act generates the same distribution of visible consequences; conversely, even a consequence that perfectly reveals the act conveys no new information when the act was already known. We call the non-instrumental preference for this conjunction a preference for *traceable agency*. The paper axiomatizes it jointly with ordinary preferences over payoff risk.

The idea has a simple formal expression. Let A be a finite set of actions, O a finite set of payoff-bearing outcomes, R a finite set of explicit visible records, and $K : A \rightarrow \Delta(O \times R)$ the channel relating actions to visible consequences. An action lottery $q \in \Delta(A)$ gives the distribution of the realised action; conditional on that action, K gives the joint distribution of payoff and record. For each fixed channel K , the primitive is a weak order $\succeq_K$ over action lotteries. Thus $q \succeq_K p$ means that q is evaluated at least as highly as p in environment K . Record processing may rewrite what is visible while leaving payoff fixed, and action processing may coarsen the description of the realised action; neither operation can increase traceability. The remaining axioms impose order, continuity, and comparison coherence, together with a dominance condition for records that arrive in stages.

The main result is an if-and-only-if characterization. A family $\{\succeq_K\}_K$ satisfies the eight axioms if and only if there are a nonconstant payoff utility $u : O \rightarrow \mathbb{R}$ and $\lambda > 0$ such that, for every finite joint channel K and every $q, p \in \Delta(A)$,

$$q \succeq_K p \iff \mathbb{E}_{q,K}[u(O)] + \lambda I_{q,K}(A; O, R) \geq \mathbb{E}_{p,K}[u(O)] + \lambda I_{p,K}(A; O, R).$$

Reversing the three trace-facing clauses yields the trace-averse dual with $\lambda < 0$; replacing them by indifference yields expected utility, or $\lambda = 0$.

The information term has a simple interpretation. Observing (O, R) induces a posterior belief about the realised action. Its traceability is the expected reduction in uncertainty about A :

$$I_{q,K}(A; O, R) = H(q) - \mathbb{E}_{q,K}[H(q(\cdot \mid O, R))].$$

If visible consequences are independent of actions, the value is zero. If the action is known in advance because q is degenerate, the value is also zero. If each action in the support of q produces a distinct visible consequence, the value is $H(q)$ —Dostoevsky’s case, in which conduct is observed directly. Thus the criterion is not a taste for randomisation as such. Randomisation matters only because it creates uncertainty about agency for the visible consequence to resolve.

A more mundane example fixes ideas. A donor can route a gift to a named project or to a pooled fund; the material effect is payoff-bearing, while the public attribution is an explicit record. Holding the distribution of material effects fixed, if the two routes generate the same distribution over records, the record is independent of the act and its additional traceability is zero, however uncertain the giving. If the routes report differently, traceability is positive exactly when the realised route is genuinely uncertain beforehand—the donor wavers, or delegates the decision—and it reaches $H(q)$ when the visible consequence identifies the route. A donor certain to give by name leaves no trace in this sense: a perfectly attributed gift that was never in doubt yields a record that only confirms what was already known. If material consequences themselves differ across routes, they too form part of the trace. The criterion prices the news a visible consequence carries about conduct, not knowledge of it.

The example ranks lotteries within a single fixed environment; the representation also speaks across environments. The environment itself is never an object of choice: the theory begins neither with a preference over pairs (q, K) nor with a numerical index on channels. When lotteries carried by different channels must be compared, the channels are embedded as labelled blocks of a single larger environment, and the comparison is an ordinary one between two block-supported lotteries in a fixed problem. The theorem represents all such comparisons by the same u and λ on one scale: it does not merely say that each environment carries a ranking ordinally equivalent to the displayed criterion; it identifies common payoff and trace units across environments.

Traceable agency cannot be reduced to any preference over consequences alone, because here the action is part of the consequence: what is valued is the joint distribution of the action with the visible pair, not the distribution of the visible pair by itself. Consider a channel with three actions, a constant payoff, and two records. The first two actions produce different records with certainty, while the third produces either record with equal probability. A fair lottery over the first two actions and a certain choice of the third induce the same joint distribution of payoff and record. Yet in the first case the record identifies the realised action, whereas in the second it conveys no information about it. Every index of the induced consequence distribution—expected utility over payoffs, or over payoff–record pairs—is indifferent, while the traceability component assigns one bit to the first lottery and zero to the second.

This difference does not make traceable agency a substitute for expected utility. The point of the present theorem is to axiomatize their joint decomposition rather than assume it:

$$V(q, K) = \mathbb{E}_{q, K}[u(O)] + \lambda I_{q, K}(A; O, R).$$

The payoff–record distinction is essential to that identification. It permits records to be processed without permitting payoff to be garbled, and it gives branchwise continuation a temporal asymmetry: records may resolve first, whereas payoff is terminal. Those restrictions derive common payoff units, the additive decomposition, and matching neutrality. No primitive payoff–trace separability axiom is imposed.

A willingness to pay for decision rights and payoff autonomy is well documented. Bartling, Fehr, and Herz infer a positive intrinsic value of decision rights for 83 percent of participants assigned to the principal role; the average control premium is 16.7 percent and significantly positive in all ten parameterisations, and an independent replication closely reproduces their individual-level results [7, 13]. Owens, Grossman, and Fackler elicit beliefs separately: expected-money maximisation predicts betting on oneself in 56.4 percent of decisions, yet subjects do so in 64.9 percent, sacrificing 8–15 percent of expected earnings [41]. Neither design varies what

consequences reveal about the act, so neither identifies the traceability channel.

Evidence closer to the action–consequence channel comes from two studies, the second incentivised, in which subjects prefer action menus with greater Jensen–Shannon divergence among consequence distributions even when current expected payoff and outcome entropy are fixed [39, 40]. Under their uniform prior over two actions, this index is exactly mutual information. Both designs, however, let token values change after the menu is chosen, so divergence also provides instrumental flexibility; a later model comparison finds that dynamic expected utility best explains the choices [37]. These studies therefore do not isolate a non-instrumental value of mutual information and, to our knowledge, no existing experiment does.

The model suggests the following direct test. Fix a channel with four actions, $a_1, \dots, a_4$. Every action yields the same monetary payoff together with a binary non-payoff record. Actions a_1 and a_2 each produce a different symbol; actions a_3 and a_4 each produce either symbol with equal probability. Participants choose between a fair coin flip over $\{a_1, a_2\}$ and one over $\{a_3, a_4\}$. The two flips induce the same joint distribution of payoff and record and involve the same degree of deliberate randomisation; only the first makes the record informative about the realised action. Traceable agency therefore predicts a strict preference for it. Because the record affects neither material payoffs nor any later decision, this comparison already excludes instrumental flexibility.¹

Let us now discuss the eight axioms. The structural axioms (A1)–(A3) provide the comparison framework. Each channel carries a continuous weak order over action lotteries. When lotteries associated with different channels must be compared, the channels are embedded as disjoint components of one larger environment; duplicating a component or adding components not involved in the comparison must not affect the ranking.

Axioms (A4) and (A5) state the two sides of data processing. The agent never gains when her record is rewritten after the fact, holding payoffs fixed, nor when her action is reported noisily. Because action processing is stated on the joint law, reported actions reached with zero probability impose no constraint: their irrelevance is a consequence of the axiom rather than a convention appended to it.

Axiom (A6) is a sure-thing principle for records that arrive in stages. If one continuation is weakly preferred after every first-stage record that can occur, then the branch-by-branch replacement cannot make the compound environment worse; a strict improvement on a positive-probability branch makes it strictly better. This condition supplies the system’s cardinal discipline. It implies invariance to common public randomisation and calibrates continuation values across posterior action lotteries.

The two orientation axioms separate payoff from trace. Axiom (A7) requires some pair of payoff outcomes to be strictly ranked. Axiom (A8) requires full revelation of a genuinely uncertain action to be strictly preferred to an uninformative record when payoff is fixed. The first makes u nonconstant; the second selects $\lambda > 0$. Neither imposes a numerical scale.

A sketch of the proof follows the theorem in Section 4; the complete argument is divided between the pure-trace proof in Appendix A and the completion of Theorem 1 in Appendix B.

2 Related Literature

The paper belongs to the decision-theoretic tradition that enriches the object of preference to capture motives that expected utility over final payoffs alone leaves out. Kreps [35] and Dekel, Lipman, and Rustichini [20] represent flexibility and subjective uncertainty about future tastes through preferences over menus, and the freedom-of-choice literature initiated by Pattanaik and Xu [42] ranks opportunity sets by the freedom they embody; here decision rights and feasible

¹A garbling control completes the design. Repeat the choice in a second treatment with a channel that writes a fair record independently of the realised action: both flips then have zero traceability, and the preference should disappear. A preference surviving the garbling would indicate salience or framing effects rather than a value of trace.

actions are held fixed, and what varies is ex-post authorship rather than ex-ante opportunity: whether the realised visible consequence statistically records the action. Gul and Pesendorfer [29] represent temptation and commitment; Kreps and Porteus [36] enrich the domain to temporal lotteries so that the timing of the resolution of uncertainty can itself carry utility; and Grant, Kajii, and Polak [28] characterise within that domain an intrinsic preference for information, valued independently of any planning use. The information valued here flows in the opposite direction—not what signals teach the agent about the world, but what visible consequences teach the world about the agent. Deliberate randomisation has also been axiomatised through a hedging motive [17]; unlike that motive, traceability depends on the observability of the action–consequence record. The nearest neighbours lie in the signaling tradition descending from Spence [48], in which actions are informative about hidden characteristics: social-image models explain attribution-seeking behaviour, such as named rather than anonymous giving, through observers’ beliefs entering utility [27, 10, 3]; self-image models direct the same inference inward, towards beliefs about one’s own latent traits [9]; and in Bernheim’s conformity model [11] agents *suppress* the legibility of their predispositions, an instrumental counterpart of the deniability orientation developed below. In all of these, what is inferred matters through its consequences—esteem, sanction, or self-regard; here no observer’s response enters utility, beliefs figure only as the statistical yardstick against which the trace is read, and the preference survives anonymity. Attribution, expressive value, and reputation are possible microfoundations for the behaviour the axioms describe, but they are not primitives of the theorem; what none of these literatures has made the object of preference is the trace itself.

The relation to axiomatic information theory is different in kind. Shannon [46] derived entropy from continuity, monotonicity on uniform distributions, and a grouping recursion; Khinchin [32] and Faddeev [24] progressively weakened these postulates, Faddeev reducing them to symmetry, continuity of the binary entropy, and the recursion that appears here as a derived object. Rényi [44] showed that relaxing the grouping postulate to general quasi-arithmetic means, while retaining additivity over independent experiments, yields the one-parameter family bearing his name, in which Shannon entropy is the unique member still satisfying the recursion; Aczél and Daróczy [1] give a systematic treatment by functional equations, and Csiszár [19] surveys the field. Mutual information itself is typically characterised indirectly, as entropy reduction or as the relative entropy between the joint law and the product of its marginals: Hobson [31] characterises relative entropy by grouping and additivity conditions, and Baez and Fritz [6], extending the treatment of entropy as a measure of information loss in Baez, Fritz, and Leinster [8], derive it as the essentially unique functorial, convex-linear, lower semicontinuous invariant of Bayesian inference.

All of these characterisations take a numerical functional as primitive and impose structural equations on distributions, random variables, or information-loss maps. The present theorem starts instead from ordinal preference relations over lotteries in fixed joint environments and derives the functional: common cardinal structure, the expected-utility component, the information chain rule, and finally Faddeev’s recursion are consequences of behavioural axioms rather than postulates on a given function. The Shannon part of the exercise is thus the information-theoretic analogue of expected-utility theory, in which the numerical form of the criterion is the output of a representation theorem rather than its input; what is characterised is not an abstract measure of information but the preference of an agent for whom that measure is the value of traceable agency.

A smaller literature takes information structures themselves as the ranked domain. Gilboa and Lehrer [26] characterise which functions on partitions are values of information for some Bayesian decision problem, and Azrieli and Lehrer [5] extend the characterisation to stochastic structures; in both, value is instrumental by construction. Cabrales, Gossner, and Serrano [14] come closest to the present representation from the instrumental side: for ruin-averse investors, the ranking of information structures is represented by entropy reduction, so that mutual

information emerges from a specific class of decision problems rather than from ordinal axioms. On the non-instrumental side, Caplin and Leahy [16] put beliefs into the utility function, and Ely, Frankel, and Kamenica [23] derive the entertainment value of suspense and surprise from preferences over the path of beliefs. The present paper combines the two strands: its domain consists of lotteries in fixed payoff–record channels, its trace valuation is non-instrumental, and, unlike either strand, the functional is not posited or induced by a decision problem but derived ordinally and pinned down jointly with material utility.

The closest formal literature axiomatises the cost of information. Pomatto, Strack, and Tamuz [43] take as primitive a cardinal cost functional on Blackwell experiments [12], defined without reference to a prior, and show that Blackwell monotonicity, continuity, additivity over independent experiments, and constant marginal cost characterise the log-likelihood ratio costs: weighted sums of Kullback–Leibler divergences between the state-conditional signal distributions. Caplin, Dean, and Leahy [15] and Denti [21] work from state-dependent stochastic choice data generated by rationally inattentive agents and characterise posterior-separable cost functions, with the Shannon cost singled out by invariance conditions; Frankel and Kamenica [25] characterise valid measures of information as expected reductions of measures of uncertainty, with entropy reduction selected by additivity requirements. Throughout this literature, information is instrumental: it concerns an exogenous state and is costly, or valuable, as an input into a downstream decision. This has two structural consequences. Blackwell monotonicity is a theorem or an unquestioned desideratum, since a decision-maker who can condition on a signal is weakly better off with a finer one; and the primitive object is cardinal, or is disciplined by an optimisation model that supplies cardinal structure.

The present paper shares the destination of that literature—coherence over mixing and sequential resolution forces the Shannon scale—but differs in primitive, in direction, and in what is derived rather than assumed. The primitive here is a family of ordinal weak orders over lotteries on the agent’s own actions; there is no downstream decision, and prior dependence is essential rather than excluded: the trace component is the whole map $(q, K) \mapsto I_{q,K}(A; O, R)$. Indeed, the two families are disjoint rather than nested: mutual information is prior-dependent and subadditive over independent experiments, so it violates the additivity that defines the log-likelihood ratio class, while fully revealing experiments, which have infinite log-likelihood ratio cost, receive the finite value $H(q)$ here. Because the primitive is ordinal and fibred by the prior, the cardinal structure that the cost literature postulates must here be constructed; branchwise continuation consistency performs that calibration jointly with the payoff scale. Finally, record and action data processing, Axioms (A4) and (A5), are primitive postulates of taste rather than theorems of instrumental rationality; as the sign trichotomy (Corollary 3) makes precise, their joint reversal is equally coherent and characterises a preference for deniability. Where the two traditions do meet—at the Shannon functional, reached from opposite ends of the channel—they axiomatise different objects: there, the price of learning about the world; here, a taste for being learned about, tempered by ordinary payoff utility.

3 The Model

Fix a finite set O of *payoff-bearing outcomes*, with $|O| \geq 2$. Let A be a nonempty finite set of *actions* and R a nonempty finite set of *explicit visible records*. A joint channel is a stochastic kernel

$$K : A \longrightarrow \Delta(O \times R).$$

For each finite joint channel K , the primitive behavioural datum is a binary relation

$$\succeq_K \quad \text{on} \quad \Delta(A).$$

The interpretation is that q is a lottery over actions used by the agent and $q \succeq_K p$ means that, in the fixed environment K , q is evaluated at least as highly as p . The lottery is the procedure

being evaluated; the channel mapping actions into payoff–record pairs is not an object of choice.

Axioms are imposed on the family $\{\succeq_K\}_K$.

(A1) Weak order. For every finite joint channel K , $\succeq_K$ is complete and transitive on $\Delta(A)$.

(A2) Continuity. For each fixed triple of alphabets (A, O, R) , the set

$$\{(K, q, p) : q \succeq_K p\} \subseteq \Delta(O \times R)^A \times \Delta(A)^2$$

is closed.

Comparison environments. The remaining axioms compare a channel with a garbling of it and a compound experiment with its branches. Each relation $\succeq_K$ ranks only lotteries, so these comparisons need a domain in which two channels can appear in one preference statement. Labelled blocks provide it, much as objective lotteries enrich Savage’s domain in Anscombe and Aumann [4], and without making the channel an object of choice.

For $K : A \rightarrow \Delta(O \times R)$ and $L : B \rightarrow \Delta(O \times S)$, let $K \sqcup L$ be the block-diagonal joint channel with action set $(A \times \{0\}) \cup (B \times \{1\})$, payoff set O , and record set $(R \times \{0\}) \cup (S \times \{1\})$, defined by

$$(K \sqcup L)(o, (r, 0) \mid (a, 0)) = K(o, r \mid a), \quad (K \sqcup L)(o, (s, 1) \mid (b, 1)) = L(o, s \mid b),$$

with all cross-block probabilities zero. For $q \in \Delta(A)$ and $p \in \Delta(B)$, let q^0 and p^1 denote their block-supported copies, defined on the tagged action set by

$$\begin{aligned} q^0(a, 0) &:= q(a), & q^0(b, 1) &:= 0, \\ p^1(a, 0) &:= 0, & p^1(b, 1) &:= p(b), \end{aligned} \quad (a \in A, b \in B).$$

When the blocks have the same action alphabet, q^0 and q^1 are the two tagged copies of the same lottery. For a finite family $\{K_i : A_i \rightarrow \Delta(O \times R_i)\}_{i \in I}$, write $\bigsqcup_{i \in I} K_i$ for the analogous block environment and q_i^i for the lottery that assigns $q_i(a)$ to (a, i) and zero to every other block. Write

$$(q, K) \succeq (p, L) \iff q^0 \succeq_{K \sqcup L} p^1.$$

Each instance of this shorthand is a single comparison inside the two-block environment $K \sqcup L$, which is itself built from the two pairs; $\succ$ and $\sim$ denote its asymmetric and symmetric parts. The pair (q, K) has no meaning in isolation.²

(A3) Block-comparison coherence. *Duplication:* for every $K : A \rightarrow \Delta(O \times R)$ and all $q, p \in \Delta(A)$,

$$q \succeq_K p \iff (q, K) \succeq (p, K).$$

²A complete numerical example. Let $A = \{a_1, a_2\}$, $B = \{b_1, b_2\}$, $O = \{o, o'\}$, and let the record alphabets be $R = \{r, r'\}$ and $S = \{s\}$. With rows indexed by actions and columns by payoff–record pairs,

$$K = \begin{array}{c|cccc} & (o, r) & (o', r) & (o, r') & (o', r') \\ \hline a_1 & 0.5 & 0 & 0 & 0.5 \\ a_2 & 0 & 0.7 & 0.3 & 0 \end{array} \quad L = \begin{array}{c|cc} & (o, s) & (o', s) \\ \hline b_1 & 0.3 & 0.7 \\ b_2 & 0.6 & 0.4 \end{array}$$

The block environment $K \sqcup L$ has tagged action set $\{(a_1, 0), (a_2, 0), (b_1, 1), (b_2, 1)\}$, payoff set O , and tagged record set $\{(r, 0), (r', 0), (s, 1)\}$; its kernel is block-diagonal, K on the first two rows, L on the last two, and every cross-block entry zero. For $q = (0.3, 0.7)$ and $p = (0.8, 0.2)$, the block-supported copies are $q^0 = (0.3, 0.7; 0, 0)$ and $p^1 = (0, 0; 0.8, 0.2)$, the semicolon separating the blocks. Then $(q, K) \succeq (p, L)$ abbreviates $q^0 \succeq_{K \sqcup L} p^1$: one comparison of two ordinary lotteries in this one environment. Both block-supported lotteries deliver o with probability 0.36, so under the representation of Theorem 1 the comparison turns on the trace terms alone—approximately 0.88 against 0.04 bits—and is strict.

Irrelevant blocks: for every finite block environment $\bigsqcup_{k \in I} K_k$, every ordered pair of distinct blocks $i, j \in I$, and all $q_i \in \Delta(A_i)$ and $q_j \in \Delta(A_j)$,

$$q_i^i \succeq_{\bigsqcup_{k \in I} K_k} q_j^j \iff (q_i, K_i) \succeq (q_j, K_j).$$

Block labels should not alter within-environment rankings, nor should unused blocks affect the comparison between active blocks.

Processing and compounding. The next three axioms use two ways of transforming an environment and one way of composing environments. A *record processor* is a stochastic kernel $T(r' \mid o, r)$ that may depend on the payoff but leaves it unchanged; it transforms $K : A \rightarrow \Delta(O \times R)$ into

$$(KT)(o, r' \mid a) := \sum_r K(o, r \mid a) T(r' \mid o, r). \quad (1)$$

An *action report* is a stochastic kernel $S : A \rightarrow \Delta(B)$; it transforms a lottery $q \in \Delta(A)$ into $(qS)(b) := \sum_a q(a) S(b \mid a)$ and K into any channel $\widehat{K} : B \rightarrow \Delta(O \times R)$ satisfying

$$(qS)(b) \widehat{K}(o, r \mid b) = \sum_a q(a) S(b \mid a) K(o, r \mid a) \quad \text{for every } (b, o, r). \quad (2)$$

If $(qS)(b) = 0$, equation (2) leaves the corresponding row of $\widehat{K}$ unrestricted, and every statement below applies to every consistent completion. Both transformations preserve the O -marginal. Finally, a first-stage record channel $P : A \rightarrow \Delta(Y)$ and a profile of continuation channels $K^y : A \rightarrow \Delta(O \times R_y)$ compose into the *compound environment*

$$(P \triangleright \{K^y\})(o, (y, r) \mid a) := P(y \mid a) K^y(o, r \mid a). \quad (3)$$

Given $q \in \Delta(A)$, a branch y is *reached* if $m(y) := \sum_a q(a) P(y \mid a) > 0$, and its branch posterior is $q_y(a) := q(a) P(y \mid a) / m(y)$.

(A4) Record data processing. For every $K : A \rightarrow \Delta(O \times R)$, every record processor T , and every $q \in \Delta(A)$,

$$(q, K) \succeq (q, KT).$$

The agent never gains when her record is garbled.

(A5) Action data processing. For every $K : A \rightarrow \Delta(O \times R)$, every action report $S : A \rightarrow \Delta(B)$, every $q \in \Delta(A)$, and every channel $\widehat{K}$ satisfying (2),

$$(q, K) \succeq (qS, \widehat{K}).$$

Nor does she gain when her action is reported noisily, holding the payoff lottery fixed.

(A6) Branchwise continuation consistency. For every finite family of record alphabets $(R_y)_{y \in Y}$, every first-stage record channel $P : A \rightarrow \Delta(Y)$, every pair of continuation profiles $K^y, L^y : A \rightarrow \Delta(O \times R_y)$, and every $q \in \Delta(A)$: if

$$(q_y, K^y) \succeq (q_y, L^y) \quad \text{for every reached } y,$$

then

$$(q, P \triangleright \{K^y\}) \succeq (q, P \triangleright \{L^y\}),$$

strictly if one reached branch comparison is strict.

The two profiles share the continuation record alphabet within each branch, although R_y may vary across branches; their channels are otherwise unrestricted. In a compound environment the branching variable is a record, and payoffs arrive only in the continuations. The axiom is a sure-thing principle across the branches of an interim record: a profile weakly better on every reached branch is weakly better overall, with unreached branches unrestricted exactly as Savage’s null events are.

(A7) Material relevance. For $o \in O$, let $K_o : \{*\} \rightarrow \Delta(O \times \{*\})$ deliver o with an uninformative record, $K_o(o, * \mid *) = 1$. For some $o^+, o^- \in O$,

$$(\delta_*, K_{o^+}) \succ (\delta_*, K_{o^-}).$$

The agent cares about material outcomes, and not only about their trace.

(A8) Positive trace orientation. For every finite A with $|A| \geq 2$, every full-support $q \in \Delta(A)$, and every $o \in O$,

$$(q, K_o^{\text{id}}) \succ (q, K_o^0),$$

where $K_o^{\text{id}}(o, a \mid a) = 1$, with record alphabet A , and $K_o^0(o, * \mid a) = 1$. At a constant payoff, a record that reveals a genuinely uncertain action is strictly better than one that reveals nothing: trace is sought, not shunned.

4 The Representation Theorem

For $q \in \Delta(A)$ and $K : A \rightarrow \Delta(O \times R)$, let

$$p_{qK}(o, r) := \sum_a q(a) K(o, r \mid a)$$

be the induced distribution of visible consequences, and define

$$I_{qK}(A; O, R) := \sum_{a, o, r} q(a) K(o, r \mid a) \log \frac{K(o, r \mid a)}{p_{qK}(o, r)}. \quad (4)$$

Summands with $q(a) K(o, r \mid a) = 0$ are set to zero, the logarithm has any fixed base, and expectations below are taken under the joint distribution $q(a) K(o, r \mid a)$.

Theorem 1 (Trace-tempered choice). *For a family $\{\succeq_K\}_K$ as described above, the following are equivalent.*

- (i) *The family satisfies Axioms (A1)–(A8).*
- (ii) *There are a nonconstant function $u : O \rightarrow \mathbb{R}$ and a number $\lambda > 0$ such that, for every nonempty finite A, R , every $K : A \rightarrow \Delta(O \times R)$, and every $q, p \in \Delta(A)$,*

$$q \succeq_K p \iff V(q, K) \geq V(p, K), \quad V(q, K) := \mathbb{E}_{q, K}[u(O)] + \lambda I_{qK}(A; O, R). \quad (5)$$

Moreover, under these equivalent conditions, block-supported cross-channel comparisons are represented on the same scale: for every finite block environment $\bigsqcup_{k \in I} K_k$, every two distinct blocks $i, j \in I$, and every $q_i \in \Delta(A_i)$ and $q_j \in \Delta(A_j)$,

$$q_i^i \succeq_{\bigsqcup_{k \in I} K_k} q_j^j \iff V(q_i, K_i) \geq V(q_j, K_j).$$

Appendix A proves the pure-trace lemma used in the argument, and Appendix B proves the equivalence and the same-scale block clause. The latter combines Axiom (A3) with the mutual-information identity for a constant block label.

Proof sketch. That the representation satisfies the axioms is a direct verification. The converse has four steps.

Step 1: reduction to terminal laws. Fix a full-support q . Two channels that induce the same joint law of realised payoff and posterior about the action are record processings of one another, so Axioms (A1), (A3), and (A4) rank them together (Lemma 35); Axiom (A5) deletes null actions, and applied to bijective reports in both directions it makes relabellings neutral. Each pair (q, K) is thereby identified with its *terminal law*, the joint distribution of realised payoff and posterior.

Step 2: cardinalisation. A public coin is a first stage that reveals nothing, so every branch posterior equals q . Axiom (A6) applied to such stages yields mixture independence over terminal laws (Lemma 35), and with continuity, Axiom (A2), the mixture-space theorem gives an affine representation on each fixed-prior fibre. All cardinal content enters here: every other axiom is an ordinal, environment-by-environment condition.

Step 3: two scales. On singleton action sets the affine representation is expected utility, nonconstant by Axiom (A7); action reports transport the payoff scale across priors and alphabets, and block coherence splices the fibre representations into one common scale (Lemma 36). On the constant-payoff restriction, Appendix A identifies the residual order with mutual information; Axiom (A8) fixes its positive orientation, and Lemma 37 puts every pure comparison on the common scale.

Step 4: assembly. Axiom (A6) enters once more: inserting continuations into a reached branch is an order isomorphism, and calibrating branches with terminal payoffs forces values to aggregate across branches at their reached probabilities (Lemmas 38 and 39). Sequentialising an arbitrary channel—announce (o, r) as a first-stage record, deliver o terminally—is neutral by Axiom (A4), and the recursion returns $\mathbb{E}_{q,K}[u(O)] + \lambda I_{q,K}(A; O, R)$.

Uniqueness To state uniqueness, fix one representation V supplied by Theorem 1. A real-valued index W on all finite lottery–channel pairs is a *common-scale representation* if, for every channel $K : A \rightarrow \Delta(O \times R)$ and every $q, p \in \Delta(A)$,

$$q \succeq_K p \iff W(q, K) \geq W(p, K),$$

and, for every finite block environment $\bigsqcup_{k \in I} K_k$, every two distinct blocks $i, j \in I$, and all $q_i \in \Delta(A_i)$ and $q_j \in \Delta(A_j)$,

$$q_i \succeq_{\bigsqcup_{k \in I} K_k} q_j^j \iff W(q_i, K_i) \geq W(q_j, K_j).$$

Call a common-scale representation W *branch-additive* if, for every first-stage record channel P , every $q \in \Delta(A)$, and every two continuation profiles $\{K^y\}$ and $\{L^y\}$ admitted in Axiom (A6),

$$W(q, P \triangleright \{K^y\}) - W(q, P \triangleright \{L^y\}) = \sum_{y: m(y) > 0} m(y) [W(q_y, K^y) - W(q_y, L^y)]. \quad (6)$$

Thus continuation-value differences aggregate by their reached-branch probabilities; no arbitrary zero for material utility is required.

Proposition 2 (Uniqueness). *Suppose the equivalent conditions of Theorem 1 hold. For any real-valued index W on all finite lottery–channel pairs:*

- (i) *W is a common-scale representation if and only if $W = \varphi \circ V$ for a single strictly increasing $\varphi : [\underline{u}, \infty) \rightarrow \mathbb{R}$, where $\underline{u} := \min_{o \in O} u(o)$.*
- (ii) *A common-scale representation W is branch-additive if and only if $W = \alpha V + \beta$ for some $\alpha > 0$ and $\beta \in \mathbb{R}$.*

The primitive remains ordinal: part (i) permits any common increasing transformation. Branch additivity is an accounting requirement on an index, not another behavioural axiom; part (ii) selects the affine gauge. Consequently, if (u, λ) and $(\tilde{u}, \tilde{\lambda})$ are two trace-tempered representations, then

$$\tilde{u} = \alpha u + \beta, \quad \tilde{\lambda} = \alpha \lambda$$

for some $\alpha > 0$ and $\beta \in \mathbb{R}$. Once two payoff outcomes are assigned numerical utilities, λ is therefore uniquely measured in those payoff units per unit of information.

Sign Uniqueness concerns scale; a complementary observation concerns sign. Axioms (A4), (A5), and (A8) orient the trace component. Write $(A4^-)$, $(A5^-)$, and $(A8^-)$ for the clauses obtained by reversing their displayed comparisons, and write $(A4^0)$, $(A5^0)$, and $(A8^0)$ for the corresponding indifference clauses.

Corollary 3 (Sign trichotomy). *Suppose Axioms (A1)–(A3), (A6), and (A7) hold. Then:*

- (i) *under Axioms (A4), (A5), and (A8), the family has the representation in Theorem 1, with $\lambda > 0$;*
- (ii) *under $(A4^-)$, $(A5^-)$, and $(A8^-)$, it has the same representation with $\lambda < 0$;*
- (iii) *under $(A4^0)$, $(A5^0)$, and $(A8^0)$, it is represented by expected utility, equivalently by the same formula with $\lambda = 0$.*

Thus the trace component has exactly the positive, negative, and neutral orientations.

The proof is in Appendix C.

For $q \in \Delta(A)$ and $K : A \rightarrow \Delta(O \times R)$, write

$$\begin{aligned} \ell_{qK}(o) &:= \sum_r p_{qK}(o, r), \\ \pi_{or}^{qK}(a) &:= \frac{q(a)K(o, r \mid a)}{p_{qK}(o, r)} \quad \text{when } p_{qK}(o, r) > 0, \\ \mu_{qK} &:= \sum_{(o, r): p_{qK}(o, r) > 0} p_{qK}(o, r) \delta_{\pi_{or}^{qK}}. \end{aligned}$$

Corollary 4 (Derived matching neutrality). *Suppose Axioms (A1)–(A8) hold. Fix A and $q \in \Delta(A)$. If channels K and L satisfy*

$$\sum_o \ell_{qK}(o)u(o) = \sum_o \ell_{qL}(o)u(o), \quad \mu_{qK} = \mu_{qL},$$

then $(q, K) \sim (q, L)$. Thus the matching of payoff labels to posterior atoms has no additional value.

The proof is in Appendix C. In words: the agent values what she obtains and what the visible consequence reveals about what she did, but not the coincidence of the two. Winning and being seen to act are priced separately; there is no premium for the good outcome arriving through the revealing record rather than alongside it. Any such premium would be a payoff–trace complementarity, and the corollary leaves no room for one. This is the behavioural face of the additive representation—the empirical content of the separability that the theorem derives rather than assumes. The separation is achieved in the domain and in Axiom (A6): compound environments may branch on interim records, never on payoffs.

Proposition 5 (Independence of the axioms). *For each $i \in \{1, \dots, 8\}$ there is a family satisfying the other seven axioms and violating Axiom (Ai).*

The eight families are constructed in Appendix C. Three mark the substantive frontier of the system. A conditional-entropy penalty satisfies every axiom except record data processing and values the removal of extraneous record noise. A Gini posterior-reduction criterion satisfies every axiom except action data processing. The satiation criterion $\mathbb{E}[u(O)] + \min\{I(A; O, R), c\}$, for $c > 0$, satisfies every axiom except branchwise continuation consistency and ceases to value trace once c units of information are secured.

Taken together, the results separate orientation, scale, and behavioural content. The sign trichotomy determines whether trace is sought, avoided, or ignored; branch additivity selects the affine gauge; and no displayed axiom is redundant. Unlike the pure-record model, the magnitude of λ is not merely a choice of information unit once payoff utility has been calibrated: it is the identified payoff–trace trade-off.

5 Choice implications

This section works out what an agent with the representation of Theorem 1 does: which finite designs falsify the model directly (§5.1), when she deliberately randomises (§5.2), what she will pay for changes to the record technology (§5.3), and what her behaviour can never disclose about the environment (§5.4). Throughout, λ is measured in payoff units per unit of information in the fixed base. Proofs for this section, together with full statements of the supporting results that the text presents in summary form, are collected in Appendix C.3; only one-line verifications are kept here.

5.1 The trace test

The functional V reads the joint distribution of the action and its visible consequence. Existing theories of choice read less: the distribution of consequences, the lottery, or both. The following experiment separates them.

In the first treatment the agent chooses between two fair coins over actions. One flips between two actions that leave distinct marks; the other flips between two actions that produce noise. Both coins yield the same payoff for certain and the same fifty–fifty distribution over records; only the first coin’s record identifies the realised action. In the second treatment the record technology is disabled: the same choice is offered in a channel where every action produces noise.

Proposition 6 (The trace test). *Four actions pay the same outcome o_0 . Two of them leave distinct marks: a_1 always produces record r_1 , and a_2 always produces the distinct record r_2 . The other two are indistinguishable: a_3 and a_4 each produce r_1 or r_2 with equal probability. Call this channel K , and let q' be the fair coin over $\{a_1, a_2\}$ and q'' the fair coin over $\{a_3, a_4\}$.*

- (i) *The two coins induce the same joint distribution of visible consequences, $p_{q'K} = p_{q''K}$; yet*

$$V(q', K) - V(q'', K) = \lambda \log 2 > 0, \quad \text{so} \quad q' \succ_K q''.$$

- (ii) *Let K° be the channel in which every action pays o_0 and produces r_1 or r_2 with equal probability. Then $q' \sim_{K^\circ} q''$.*

The prediction is strict preference for the revealing coin in the first treatment and exact indifference in the second. Part (i) alone rules out every consequentialist theory. The two coins induce the same distribution over payoff–record pairs, so any criterion that evaluates only that distribution is indifferent between them: expected utility over payoffs, expected utility that values records directly as consequences through some $v(o, r)$, and every non-linear functional of $p_{q'K}$. What separates the coins is the joint distribution of the action with its visible consequence, and that is precisely what consequentialist theories do not read.

Part (i) restricts only how a theory reads consequences. A theory that also reads the lottery can match part (i). Existing theories of deliberate randomisation are of this kind: in the hedging model of Cerreia-Vioglio, Dillenberger, Ortoleva, and Riella [17], and in the preference for randomisation documented by Agranov and Ortoleva [2], the value of mixing depends only on the induced distribution and the lottery. Call an index W *channel-blind* if it reads only these two objects: there are functions F and $h : \Delta(A) \rightarrow \mathbb{R}$, fixed across channels, such that

$$W(q, K) = F(p_{qK}, h(q)).$$

The second treatment rules these out. Nothing a channel-blind index reads changes between the treatments: in all four lottery–channel combinations the distribution of visible consequences is uniform on $\{(o_0, r_1), (o_0, r_2)\}$, and the lotteries are the same two lotteries. A channel-blind index therefore separates the coins in both treatments or in neither, so none reproduces the pattern of Proposition 6: strict preference under K and indifference under K° .³

The test also rejects. The first treatment separates the three cases of Corollary 3: the revealing coin if $\lambda > 0$, the other coin if $\lambda < 0$, indifference if $\lambda = 0$. In the second treatment every representation assigns both coins the same value, whatever u and λ , so a strict preference there rejects the model.

5.2 Deliberate randomisation

A degenerate lottery leaves no trace, so an agent with $\lambda > 0$ may deliberately randomise: she gives up expected payoff to make her act genuinely uncertain, because only then can its consequence reveal it. In a payoff-based theory a sufficiently large payoff advantage makes a single action optimal. Here it need not: an action that leaves a mark no other action can leave is never wholly abandoned, however poorly it pays, because the first grain of mass on it is infinitely informative per unit, and no finite payoff difference outweighs that. The next proposition states this, and gives the exact terms on which a pure action survives: its payoff advantage must cover, for every rival, the price of the rival’s distinctiveness from it.

Some notation. Let $Z := O \times R$ and write $K_Z(z | a) := K(o, r | a)$ for $z = (o, r)$; let

$$\bar{u}(a) := \sum_{o,r} K(o, r | a) u(o), \quad m_{q,K}(z) := \sum_a q(a) K_Z(z | a)$$

be the expected payoff of an action and the distribution of visible consequences induced by a lottery. The Kullback–Leibler divergence $D_{\text{KL}}(P \| Q) := \sum_z P(z) \log(P(z)/Q(z))$ between distributions on Z measures how sharply consequences drawn from P stand out against a background of Q ; it equals $+\infty$ when P puts mass where Q does not.

Proposition 7 (Exercised distinctiveness). *Let $Z_K := \{z \in Z : K_Z(z | a) > 0 \text{ for some } a \in A\}$.*

(i) *Every optimal lottery q^* satisfies $m_{q^*,K}(z) > 0$ for every $z \in Z_K$. Consequently, if every action a has a private consequence—some z_a with $K_Z(z_a | a) > 0$ and $K_Z(z_a | b) = 0$ for all $b \neq a$ —then every optimal lottery has full support.*

(ii) *The pure lottery δ_a is optimal if and only if*

$$\bar{u}(b) + \lambda D_{\text{KL}}(K_Z(\cdot | b) \| K_Z(\cdot | a)) \leq \bar{u}(a) \quad \text{for every } b \in A,$$

and uniquely optimal if these inequalities are strict for $b \neq a$. In particular, if some rival reaches a consequence outside the support of $K_Z(\cdot | a)$, then δ_a is not optimal, however large its payoff advantage.

³If h does not depend on how the actions are labelled—an entropy bonus, say—the first treatment is enough: the permutation that swaps a_1 with a_3 and a_2 with a_4 turns q' into q'' and leaves p_{qK} unchanged, so $W(q', K) = W(q'', K)$. The second treatment is needed for arbitrary h .

With two actions, the threshold below which the agent mixes is a case of Proposition 7(ii). Let $A = \{a, b\}$, write $K_a := K_Z(\cdot \mid a)$ and $K_b := K_Z(\cdot \mid b)$, and let $\Delta := \bar{u}(a) - \bar{u}(b) \geq 0$ be the payoff advantage of a . Then δ_a is optimal if and only if $\Delta \geq \lambda D_{\text{KL}}(K_b \parallel K_a)$. The threshold is directional: what protects the pure action is the detectability of the rival against the background of its own signature. It is infinite when K_b puts mass where K_a does not—the two-action face of Proposition 7(i).

Below the threshold the agent mixes (Proposition 40, Appendix C.3). The optimal mix is unique, and it puts more mass on the inferior action the larger is λ and less the larger is Δ . Mixing therefore stops exactly at $\Delta = \lambda D_{\text{KL}}(K_b \parallel K_a)$, and since the divergence is known once the channel is designed, a single switching experiment identifies λ . At equal payoffs the mix is the capacity-achieving input of the channel, whatever λ : an agent who values trace at all, however little, randomises at zero stakes so as to maximise the information the visible consequence carries about her act. The comparative statics restate the wedge of the trace test (§5.1): randomisation here is instrumental to trace, so it strengthens with the informativeness of the channel and vanishes on uninformative ones, whereas a taste for mixing as such is channel-blind.

With more than two actions, the same logic is captured by a first-order condition. The objective $q \mapsto V(q, K)$ is concave in q , so an optimum exists and Kuhn–Tucker conditions characterise it: at an optimum, every action in use offers the same total of expected payoff and priced distinctiveness.

Proposition 8 (Distinctiveness condition). *Let $\lambda > 0$. A lottery q^* is optimal if and only if there is a constant c such that*

$$\bar{u}(a) + \lambda D_{\text{KL}}(K_Z(\cdot \mid a) \parallel m_{q^*, K}) = c \quad \text{for } a \in \text{supp}(q^*), \quad \leq c \quad \text{for } a \notin \text{supp}(q^*).$$

The proposition gives a *distinctiveness premium*. Each action is credited, on top of its expected payoff, with λ times how far its visible consequences depart from the average visible behaviour that the lottery itself generates; at an optimum this total is equalised across the actions in use. An action can therefore receive more probability because it pays more or because its consequences are more distinctive. The condition mirrors rational-inattention first-order conditions with the channel direction and sign reversed [47, 38].

Which lotteries are optimal depends on the coupling of actions to consequences, not on consequences alone; and optimal lotteries need not be unique. Yet every optimal lottery induces the same distribution of visible consequences. The model therefore predicts the record an observer sees, not the mix that generated it.

Proposition 9 (Optimal marginal and multiplicity). *Write $P_a := K_Z(\cdot \mid a)$ and $C^*(K) := \max_{q \in \Delta(A)} V(q, K)$. There is a unique $m^* \in \Delta(Z)$ such that $m_{q^*, K} = m^*$ for every optimal lottery q^* . With*

$$\mathcal{A}^* := \{a \in A : \bar{u}(a) + \lambda D_{\text{KL}}(P_a \parallel m^*) = C^*(K)\},$$

the set of optimal lotteries is exactly

$$\{q \in \Delta(A) : m_{q, K} = m^*, \text{ supp}(q) \subseteq \mathcal{A}^*\},$$

and the optimum is unique whenever the rows $\{P_a : a \in \mathcal{A}^\}$ are affinely independent. If actions are partitioned into classes with identical rows, $V(q, K)$ depends on q within each class only through the class's total mass: every redistribution of an optimal lottery within such a class is optimal, and the optimal lotteries are exactly the lifts of the optimal lotteries of the channel that collapses the classes.*

The proposition's final claim—that moving mass between actions with identical rows changes nothing—is a test. Split an action into two copies with the same row. The record cannot tell the copies apart, so the value, the induced marginal m^* , and the pair's total probability do not

depend on how the mass is divided between them. Any model that gives each action its own weight, as multinomial logit does, predicts instead that duplication raises the pair’s combined share. Trace-tempered choice predicts exact invariance.

Remark 10 (Logit boundary). Suppose every action leaves its own mark: the supports of the rows are pairwise disjoint, so the visible consequence identifies the action and $I_{qK}(A; O, R) = H(q)$. The unique optimum is then multinomial logit,

$$q^*(a) = \frac{\exp(\bar{u}(a)/\lambda)}{\sum_{a'} \exp(\bar{u}(a')/\lambda)},$$

with $1/\lambda$ as the payoff sensitivity (information in natural logarithms; a change of base rescales λ). If instead no action leaves any mark—identical rows—every lottery is optimal. Between these extremes there is no closed form, and the ratio of two actions’ probabilities depends on the rest of the menu: by Proposition 8, each action is credited for standing out against the average consequence of the mix, and adding an action changes that average. So independence of irrelevant alternatives holds when every action leaves its own mark and fails otherwise—and it fails in the direction the red-bus/blue-bus critique demands, since an action that resembles its rivals leaves a less distinctive mark. Failures of independence in choice data are commonly attributed to correlated tastes or to costly attention. A taste for trace produces them as well. In rational inattention the same logit formula arises for the opposite reason, as the optimal tilting of a chosen channel at a fixed prior over states [38]; here the channel is fixed and the source is chosen.

5.3 The price of being on the record

Because Theorem 1 represents comparisons across environments by one V , the model prices not only lotteries but changes to the record technology itself: what is it worth to the agent that her coin is observed, that her record is coarsened, that a further signal about her is disclosed? Each answer below is an exact amount of information multiplied by λ , and each corresponds to a block comparison in the primitive domain. The first result concerns the mixing device.

Corollary 11 (The recorded coin). *Let $K, L : A \rightarrow \Delta(O \times R)$ and $t \in (0, 1)$. Let $tK \oplus (1 - t)L$ be the public mixture that draws an action-independent coin Y with $\Pr(Y = 0) = t$, applies K if $Y = 0$ and L otherwise, and includes the coin in the record; let $M_t := tK + (1 - t)L$ be the rowwise mixture, which hides it. Then, for every $q \in \Delta(A)$,*

- (i) $V(q, tK \oplus (1 - t)L) = tV(q, K) + (1 - t)V(q, L)$;
- (ii) $V(q, tK \oplus (1 - t)L) - V(q, M_t) = \lambda I(A; Y \mid O, R) \geq 0$, with equality if and only if A and Y are conditionally independent given (O, R) .

The proof is in Appendix C.

Part (i) is the value-level face of derived public-coin independence (Lemma 35): the agent has no intrinsic taste for or against randomising over environments, so a publicly randomised environment is never ranked strictly above or below both of its constituents. Part (ii) says what she does care about: whether the coin is on the record. Hiding the randomisation device costs $\lambda I(A; Y \mid O, R)$, which is strictly positive whenever knowing the operating regime would sharpen inference about the action given the visible consequences—even though the coin is independent of the action ex ante. A trace-tempered agent weakly prefers her mixing device publicised; theories that attach value to randomisation as such are silent on this visibility margin.

Behind part (ii) is the exact price of any record garbling. Axiom (A4) asserts only a weak preference against garbling; the representation upgrades the axiom to an exact price and characterises when the price is zero.

Proposition 12 (Price of a record garbling). *Let $K : A \rightarrow \Delta(O \times R)$, let $T(r' | o, r)$ be a record processor as in (1), and write $L := KT$. Fix $q \in \Delta(A)$ and let (A, O, R, R') carry the joint law $q(a)K(o, r | a)T(r' | o, r)$. Then*

$$V(q, K) - V(q, L) = \lambda I(A; R | O, R') \geq 0,$$

and the following are equivalent: (a) $V(q, L) = V(q, K)$; (b) A and (O, R) are conditionally independent given (O, R') ; (c) $\mu_{qL} = \mu_{qK}$.

Rewriting the record costs λ times the information about the action that the original record retains beyond the rewritten one, and a garbling is exactly free when it pools only visible consequences that already induced the same posterior about the action: an agent with a trace motive does not pay for gratuitous detail, only for detail that discriminates among her own actions. Symmetrically, appending to the record any further signal S drawn given (a, o, r) raises $V(q, \cdot)$ by exactly $\lambda I(A; S | O, R) \geq 0$: more evidence about oneself never hurts, a conclusion the deniability dual of Corollary 3 reverses sign for sign. Since the visible pair always includes the payoff coordinate, the total value of trace in any environment is bounded by $\lambda \min\{H(q), \log(|O| |R|)\}$.

These prices invite measurement: how much payoff would the agent give up for a more revealing record? The difficulty is that the domain has no money—the finite outcome set contains no outcome worth $u(o) + \tau$ —so a compensating offset must be manufactured from the two outcomes that Axiom (A7) strictly ranks, using comparisons that stay inside the primitive domain. Let o^+ and o^- attain the maximum and minimum of u , write $\Delta u := u(o^+) - u(o^-) > 0$, and for $\beta \in [0, 1]$ let $w(\beta) := \beta u(o^+) + (1 - \beta)u(o^-)$.

Definition 13 (Sweetened dilution). For $K : A \rightarrow \Delta(O \times R)$, $\varepsilon \in [0, 1]$, and $\beta \in [0, 1]$, let $P_\varepsilon : A \rightarrow \Delta(\{0, 1\})$ be the public coin $P_\varepsilon(1 | a) = \varepsilon$, and let $B_\beta : A \rightarrow \Delta(O \times \{*\})$ have the constant rows $B_\beta(o^+, * | a) = \beta$ and $B_\beta(o^-, * | a) = 1 - \beta$. The (ε, β) -sweetened dilution of K is the compound channel

$$D_{\varepsilon, \beta}K := P_\varepsilon \triangleright \{K, B_\beta\},$$

as in (3), listing the branch-0 and branch-1 continuations in order.

Proposition 14 (Willingness to pay for trace). *Fix $q \in \Delta(A)$, $p \in \Delta(B)$, and channels $K : A \rightarrow \Delta(O \times R)$ and $L : B \rightarrow \Delta(O \times S)$.*

(i) *For every $\varepsilon \in [0, 1]$ and $\beta \in [0, 1]$,*

$$V(q, D_{\varepsilon, \beta}K) = (1 - \varepsilon)V(q, K) + \varepsilon w(\beta).$$

(ii) *If $\varepsilon \in (0, 1)$ and $\beta, \beta' \in [0, 1]$ satisfy $(q, D_{\varepsilon, \beta}K) \sim (p, D_{\varepsilon, \beta'}L)$, then*

$$\frac{\varepsilon}{1 - \varepsilon} [w(\beta) - w(\beta')] = V(p, L) - V(q, K).$$

When the material parts agree, $\mathbb{E}_{q, K}[u(O)] = \mathbb{E}_{p, L}[u(O)]$, the elicited offset equals $\lambda[I_{pL}(B; O, S) - I_{qK}(A; O, R)]$: information is priced linearly at rate λ .

(iii) *Such a pair (β, β') exists if and only if $|V(p, L) - V(q, K)| \leq \frac{\varepsilon}{1 - \varepsilon} \Delta u$; a boundary response therefore brackets the value difference instead of measuring it.*

The public coin is incentive-neutral: it rescales both blocks' values identically, so the elicited odds gap measures only the value difference, whatever the dilution. The price it recovers is, moreover, unique and portable.

Proposition 15 (Single information price). *Write $E_{qK} := \mathbb{E}_{q, K}[u(O)]$ and $\phi(q, K) := (I_{qK}(A; O, R), E_{qK})$, and let $\underline{u} := \min_o u(o)$ and $\bar{u} := \max_o u(o)$.*

- (i) The image of ϕ over all finite pairs (q, K) is exactly $[0, \infty) \times [\underline{u}, \bar{u}]$, and the indifference classes of the block-comparison relation are, in these coordinates, the nonempty intersections of the lines $E = c - \lambda I$ with that strip.
- (ii) If $(q, K) \sim (p, L)$ and $I_{qK} \neq I_{pL}$, then

$$\frac{E_{pL} - E_{qK}}{I_{qK} - I_{pL}} = \lambda,$$

and λ is the unique number with this property: for every $\lambda' \neq \lambda$ there are finite pairs $(q, K) \sim (p, L)$ with $E_{qK} + \lambda' I_{qK} \neq E_{pL} + \lambda' I_{pL}$.

At any fixed calibration of payoff units, the cross-environment indifference map in the information–payoff plane consists of parallel straight lines of slope $-\lambda$: bits carry one price, the same at every prior, every information level, and in every finite environment, as Proposition 2 transports it. This is the sharpest testable fingerprint of branchwise continuation consistency. The satiation criterion of Proposition 5, which fails exactly Axiom (A6), produces kinked indifference curves whose measured slope collapses to zero beyond the cap; constancy of the elicited slope across designs with different priors or alphabets is therefore a direct over-identification test. Appendix C.3 records a complete two-treatment protocol implementing this elicitation (Remark 43), with the information gap fixed by design and the garbling control built in.

Choices, and not only indifferences, bound the same price. Two behaviours ordered by λ are ordered in what they choose: a higher weight buys more information and less payoff, and the rate of exchange between the two brackets the weight itself.

Remark 16 (Trace demand). Fix K , write $e(q) := \mathbb{E}_{q,K}[u(O)]$ and $i(q) := I_{qK}(A; O, R)$, and let q_j maximise $e + \lambda_j i$ for $0 < \lambda_1 < \lambda_2$. Comparing each optimum at the other weight gives $i_1 \leq i_2$, $e_1 \geq e_2$, and, whenever behaviour responds,

$$\lambda_1 \leq \frac{e_1 - e_2}{i_2 - i_1} \leq \lambda_2 :$$

the payoff surrendered per additional unit of information brackets the trace weight, whatever selection among optima the data reflect. The indirect value is convex in λ , with the chosen information as its derivative wherever it is differentiable, and its extreme regimes are expected utility ($\lambda \downarrow 0$, with information breaking material ties) and maximal distinctiveness ($\lambda \uparrow \infty$, with payoff breaking information ties). Exact statements are Proposition 41 and Corollary 42 in Appendix C.3.

5.4 What preferences cannot reveal

Suppose an agent ranks environment K above environment L at *every* lottery. Has she thereby revealed that L is a coarsening of K —that some garbling maps the one record technology into the other? For a Blackwell decision-maker the answer would be yes: dominance in every decision problem is equivalent to garbling [12]. Here the answer is no, and the gap can be stated exactly.

Definition 17. Let $K : A \rightarrow \Delta(O \times R)$ and $L : A \rightarrow \Delta(O \times R')$ have identical payoff kernels, $\sum_r K(o, r \mid a) = \sum_{r'} L(o, r' \mid a)$ for every (a, o) . Write $K \succeq_g L$ if $L = KT$ for some record processor T as in (1), and $K \succeq_c L$ if $I_{qK}(A; O, R) \geq I_{qL}(A; O, R')$ for every $q \in \Delta(A)$; the latter is the *more capable* order on the visible-consequence channels [34].

Proposition 18 (Uniform dominance is more-capable, not garbling). *Let K and L have identical payoff kernels.*

- (i) $K \succeq_g L \Rightarrow K \succeq_c L$, and $K \succeq_c L$ holds if and only if $V(q, K) \geq V(q, L)$ for every $q \in \Delta(A)$.

- (ii) *The converse of the first implication fails: there are channels with identical payoff kernels such that $V(q, K) \geq V(q, L)$ for every q and every representation (u, λ) , yet $L \neq KT$ for every record processor T .*

The reason Blackwell’s equivalence fails here is that his test family is rich—all state-dependent utilities—while the axioms pin the attitude to records down to the single functional $\mathbb{E}[u] + \lambda I$, so environments are assessed only through mutual information at priors. The revealed informativeness order is then exactly the information theorists’ more-capable order, strictly coarser than garbling. And the counterexample is utility-robust—the payoff terms cancel identically—so observing that an agent always ranks one environment above another licenses no inference that the second is a censored version of the first.

6 Interpretation and scope

Evidence, not authorship. The name promises agency; the functional rewards uncertainty. An agent certain of her act earns zero trace value, however perfectly the record attributes it, while one who hands the decision to a fair coin earns the most the channel allows. We concede what this invites: the theory prices evidence—the news a visible consequence carries about conduct—and news requires antecedent doubt, which is Dostoevsky’s own logic. The zero at certainty is a discriminating prediction rather than an embarrassment. Image models value the level of an observer’s regard, so anticipated attribution keeps its worth for an agent certain of her act; here trace dies with the doubt. Two readings of q are consistent with this: a mixing device the agent operates, which makes the randomisation results of Section 5 literal, or an observer’s prior or population frequency against which a realised act is evaluated. Under the second reading q is exogenous, and the theorem is silent on how the observer forms it.

Trace or control. Mutual information is symmetric, and the domain inherits the symmetry. A taste for visible consequences that reveal the act is behaviourally indistinguishable from a taste for acts that statistically shape visible consequences: the theorem pins down the value of the coupling between action and consequence, and trace and control are two readings of that one object. No data in a channel-indexed domain can separate them, much as Savage’s data cannot separate belief from state-dependent taste. Separation would require interventional primitives—an audience that can be removed while the channel is left intact—and these lie outside the theory. Every choice implication in Section 5 therefore tests both readings at once.

Empowerment. At constant material utility, maximising V yields the Shannon capacity of the visible-consequence channel: the empowerment index of artificial-intelligence and complex-systems research [33, 45]. The connection is stronger than a shared optimum. The theorem gives a preference foundation for the entire map $(q, K) \mapsto I_{q,K}$, so it covers constrained behaviour and choices tempered by material payoffs, and it identifies λ , the payoff price of a unit of empowerment. The connection is also narrow in an instructive way: the criterion depends only on actions assigned positive probability, so it captures control that is exercised, not options held in reserve—a genuine option value would require preferences over menus, outside this domain. Each side gains something: agency research acquires a behavioural test for its objective, and decision theory acquires an objective already in use in another field.

A belief-based utility, located. At a fixed prior the trace term is a belief-based utility: the expected reduction in the entropy of the posterior about the act, a posterior-separable felicity in the manner of [16]—though the belief concerns the agent’s own conduct, not an external state. What the axioms add is the location of the functional form in behaviour. Entropy is nowhere assumed; branchwise continuation consistency carries the whole cardinal content,

and it is a substantive claim, not a convenience. Its fingerprint is that bits carry one price: cross-environment indifference curves in the information–payoff plane are parallel straight lines (Proposition 15). An agent whose taste for trace satiates violates exactly this axiom, produces kinked curves, and satisfies every other. The theorem’s scope is agents for whom that constancy of price holds.

One-shot, with no audience in the model. Beliefs enter only as the yardstick for how much the realised consequence narrows uncertainty about the action. No observer responds: the representation predicts neither praise nor sanction nor any return to reputation, and the staged records of Axiom (A6) introduce sequencing of evidence, not repeated interaction. This is the right first step, not an oversight: it isolates the intrinsic value of trace, so that a strategic or dynamic extension can add an audience’s response without confusing it with the motive identified here.

The split is scaffolding. The payoff–record split is scaffolding, not substance. Identification needs two independently variable directions and a temporal order: records may be garbled while payoff stays intact, and terminal payoff anchors a common scale. But the object valued is the undivided pair; the theorem separates O from R only to recombine them in $I(A; O, R)$. The Underground Man, on this reading, was not asking to be rid of reason: he was asking for a lottery genuinely open beforehand and a consequence able to tell his acts apart, and the cakes he stood ready to risk are what λ measures.

References

- [1] János Aczél and Zoltán Daróczy. *On Measures of Information and Their Characterizations*. Academic Press, New York, 1975.
- [2] Marina Agranov and Pietro Ortoleva. Stochastic choice and preferences for randomization. *Journal of Political Economy*, 125(1):40–68, 2017.
- [3] James Andreoni and B. Douglas Bernheim. Social image and the 50–50 norm: A theoretical and experimental analysis of audience effects. *Econometrica*, 77(5):1607–1636, 2009.
- [4] Francis J. Anscombe and Robert J. Aumann. A definition of subjective probability. *Annals of Mathematical Statistics*, 34(1):199–205, 1963.
- [5] Yaron Azrieli and Ehud Lehrer. The value of a stochastic information structure. *Games and Economic Behavior*, 63(2):679–693, 2008.
- [6] John C. Baez and Tobias Fritz. A Bayesian characterization of relative entropy. *Theory and Applications of Categories*, 29(16):421–456, 2014.
- [7] Björn Bartling, Ernst Fehr, and Holger Herz. The intrinsic value of decision rights. *Econometrica*, 82(6):2005–2039, 2014.
- [8] John C. Baez, Tobias Fritz, and Tom Leinster. A characterization of entropy in terms of information loss. *Entropy*, 13(11):1945–1957, 2011.
- [9] Roland Bénabou and Jean Tirole. Self-confidence and personal motivation. *Quarterly Journal of Economics*, 117(3):871–915, 2002.
- [10] Roland Bénabou and Jean Tirole. Incentives and prosocial behavior. *American Economic Review*, 96(5):1652–1678, 2006.

- [11] B. Douglas Bernheim. A theory of conformity. *Journal of Political Economy*, 102(5):841–877, 1994.
- [12] David Blackwell. Equivalent comparisons of experiments. *Annals of Mathematical Statistics*, 24(2):265–272, 1953.
- [13] Justin Buffat, Matthias Praxmarer, and Matthias Sutter. The intrinsic value of decision rights: A replication and an extension to team decision making. *Journal of Economic Behavior & Organization*, 209:560–571, 2023.
- [14] Antonio Cabrales, Olivier Gossner, and Roberto Serrano. Entropy and the value of information for investors. *American Economic Review*, 103(1):360–377, 2013.
- [15] Andrew Caplin, Mark Dean, and John Leahy. Rationally inattentive behavior: Characterizing and generalizing Shannon entropy. *Journal of Political Economy*, 130(6):1676–1715, 2022.
- [16] Andrew Caplin and John Leahy. Psychological expected utility theory and anticipatory feelings. *Quarterly Journal of Economics*, 116(1):55–79, 2001.
- [17] Simone Cerreia-Vioglio, David Dillenberger, Pietro Ortoleva, and Gil Riella. Deliberately stochastic. *American Economic Review*, 109(7):2425–2445, 2019.
- [18] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley-Interscience, Hoboken, second edition, 2006.
- [19] Imre Csiszár. Axiomatic characterizations of information measures. *Entropy*, 10(3):261–273, 2008.
- [20] Eddie Dekel, Barton L. Lipman, and Aldo Rustichini. Representing preferences with a unique subjective state space. *Econometrica*, 69(4):891–934, 2001.
- [21] Tommaso Denti. Posterior separable cost of information. *American Economic Review*, 112(10):3215–3259, 2022.
- [22] Fyodor Dostoevsky. *Notes from the Underground*, Part I. 1864. Translated by Constance Garnett, 1918.
- [23] Jeffrey Ely, Alexander Frankel, and Emir Kamenica. Suspense and surprise. *Journal of Political Economy*, 123(1):215–260, 2015.
- [24] D. K. Faddeev. On the concept of entropy of a finite probabilistic scheme. *Uspekhi Matematicheskikh Nauk*, 11(1):227–231, 1956. In Russian.
- [25] Alexander Frankel and Emir Kamenica. Quantifying information and uncertainty. *American Economic Review*, 109(10):3650–3680, 2019.
- [26] Itzhak Gilboa and Ehud Lehrer. The value of information—an axiomatic approach. *Journal of Mathematical Economics*, 20(5):443–459, 1991.
- [27] Amihai Glazer and Kai A. Konrad. A signaling explanation for charity. *American Economic Review*, 86(4):1019–1028, 1996.
- [28] Simon Grant, Atsushi Kajii, and Ben Polak. Intrinsic preference for information. *Journal of Economic Theory*, 83(2):233–259, 1998.
- [29] Faruk Gul and Wolfgang Pesendorfer. Temptation and self-control. *Econometrica*, 69(6):1403–1435, 2001.

- [30] I. N. Herstein and John Milnor. An axiomatic approach to measurable utility. *Econometrica*, 21(2):291–297, 1953.
- [31] Arthur Hobson. A new theorem of information theory. *Journal of Statistical Physics*, 1(3):383–391, 1969.
- [32] Aleksandr I. Khinchin. *Mathematical Foundations of Information Theory*. Dover, New York, 1957.
- [33] Alexander S. Klyubin, Daniel Polani, and Chrystopher L. Nehaniv. Empowerment: A universal agent-centric measure of control. In *Proceedings of the 2005 IEEE Congress on Evolutionary Computation*, volume 1, pages 128–135. IEEE, 2005.
- [34] János Körner and Katalin Marton. Comparison of two noisy channels. In Imre Csiszár and Peter Elias, editors, *Topics in Information Theory*, Colloquia Mathematica Societatis János Bolyai, volume 16, pages 411–423. North-Holland, Amsterdam, 1977.
- [35] David M. Kreps. A representation theorem for preference for flexibility. *Econometrica*, 47(3):565–577, 1979.
- [36] David M. Kreps and Evan L. Porteus. Temporal resolution of uncertainty and dynamic choice theory. *Econometrica*, 46(1):185–200, 1978.
- [37] Mimi Liljeholm. Flexible control as surrogate reward or dynamic reward maximization. *Cognition*, 229:105262, 2022.
- [38] Filip Matějka and Alisdair McKay. Rational inattention to discrete choices: A new foundation for the multinomial logit model. *American Economic Review*, 105(1):272–298, 2015.
- [39] Prachi Mistry and Mimi Liljeholm. Instrumental divergence and the value of control. *Scientific Reports*, 6:36295, 2016.
- [40] Kaitlyn G. Norton and Mimi Liljeholm. The rostrolateral prefrontal cortex mediates a preference for high-agency environments. *Journal of Neuroscience*, 40(22):4401–4409, 2020.
- [41] David Owens, Zachary Grossman, and Ryan Fackler. The control premium: A preference for payoff autonomy. *American Economic Journal: Microeconomics*, 6(4):138–161, 2014.
- [42] Prasanta K. Pattanaik and Yongsheng Xu. On ranking opportunity sets in terms of freedom of choice. *Recherches Économiques de Louvain*, 56(3–4):383–390, 1990.
- [43] Luciano Pomatto, Philipp Strack, and Omer Tamuz. The cost of information: The case of constant marginal costs. *American Economic Review*, 113(5):1360–1393, 2023.
- [44] Alfréd Rényi. On measures of entropy and information. In *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability*, volume 1, pages 547–561. University of California Press, 1961.
- [45] Christoph Salge, Cornelius Glackin, and Daniel Polani. Empowerment—an introduction. In Mikhail Prokopenko, editor, *Guided Self-Organization: Inception*, pages 67–114. Springer, Berlin, 2014.
- [46] Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27:379–423, 623–656, 1948.
- [47] Christopher A. Sims. Implications of rational inattention. *Journal of Monetary Economics*, 50(3):665–690, 2003.

- [48] Michael Spence. Job market signaling. *Quarterly Journal of Economics*, 87(3):355–374, 1973.
- [49] Aaron D. Wyner and Jacob Ziv. A theorem on the entropy of certain binary sequences and applications: Part I. *IEEE Transactions on Information Theory*, 19(6):769–772, 1973.

Proofs

A Pure trace on the constant-payoff domain

Fix an outcome $o_0 \in O$. Every channel between nonempty finite sets $P : A \rightarrow \Delta(X)$ has the constant-payoff lift

$$K_P(o, x \mid a) := \mathbf{1}_{\{o=o_0\}} P(x \mid a). \quad (7)$$

Throughout this appendix, comparisons of pure record channels mean comparisons of these lifts in the model already defined in the main text:

$$q \succeq_P q' \iff q \succeq_{K_P} q'.$$

We similarly write $(q, P) \succeq (r, Q)$ for the main-text two-block comparison of (q, K_P) and (r, K_Q) . Likewise, $P \sqcup Q$, block-supported lotteries, action and record processing, and a compound $P_1 \triangleright \{Q^x\}$ are shorthand for the corresponding main-text constructions after applying (7). The lift commutes with each construction up to canonical tagged or singleton-record bijections; those bijections are neutral by Axiom (A4) applied in both directions. No new primitive relation or behavioural condition is introduced here.

Two pieces of notation will be used repeatedly. If $S : A \rightarrow \Delta(B)$ is an action report, let $S^q P : B \rightarrow \Delta(X)$ denote any Bayesian pushforward satisfying

$$(qS)(b)(S^q P)(x \mid b) = \sum_a q(a) S(b \mid a) P(x \mid a).$$

Its rows at zero-mass reports are arbitrary, exactly as in (2). For channels $P : A \rightarrow \Delta(X)$ and $Q : B \rightarrow \Delta(Y)$, their independent product is

$$(P \otimes Q)((x, y) \mid (a, b)) := P(x \mid a) Q(y \mid b).$$

For $q \in \Delta(A)$ and $P : A \rightarrow \Delta(X)$, write

$$m_{q,P}(x) := \sum_a q(a) P(x \mid a), \quad r_x^{q,P}(a) := \frac{q(a) P(x \mid a)}{m_{q,P}(x)} \quad \text{when } m_{q,P}(x) > 0,$$

and define the finite posterior law

$$\mu_{q,P} := \sum_{x: m_{q,P}(x) > 0} m_{q,P}(x) \delta_{r_x^{q,P}}.$$

Let Id_A be full revelation and U_A the one-record uninformative channel. With logarithms in the fixed base used in the main text, put

$$I_{q,P}(A; X) := H(m_{q,P}) - \sum_a q(a) H(P(\cdot \mid a)) = H(q) - \int_{\Delta(A)} H(r) d\mu_{q,P}(r).$$

Proposition 19 (Pure-trace lemma). *Suppose the preference family satisfies main-text Axioms (A1)–(A6) and (A8). Then, for every record channel between nonempty finite sets $P : A \rightarrow \Delta(X)$ and every $q, q' \in \Delta(A)$,*

$$q \succeq_P q' \iff I_{q,P}(A; X) \geq I_{q',P}(A; X).$$

The same numerical scale represents every block-supported comparison: for every nonempty finite family $\{P_i : A_i \rightarrow \Delta(X_i)\}_{i \in J}$ of channels between nonempty finite sets, distinct $i, j \in J$, and priors $q_i \in \Delta(A_i)$ and $q_j \in \Delta(A_j)$,

$$q_i^i \succeq_{\bigsqcup_{k \in J} P_k} q_j^j \iff I_{q_i, P_i}(A_i; X_i) \geq I_{q_j, P_j}(A_j; X_j).$$

A.1 Proof

Assume the hypotheses of Proposition 19. Every appeal below to weak order, continuity, block coherence, record processing, action processing, branchwise continuation, or positive trace orientation is an appeal to Axiom (A1), (A2), (A3), (A4), (A5), (A6), or (A8), respectively, on the constant-payoff lifts. Thus the proof has no auxiliary axiom system.

How to read the proof. The argument has five interfaces. Each stage ends with exactly the object needed by the next one:

- Stage 1: replace channels by Bayes-plausible posterior laws, derive public-mixture independence, and cardinalise each fixed-prior quotient using only the segment closedness supplied by primitive continuity;
- Stage 2: make relabelling, undetectable distinctions, and restriction to the positive support exact consequences of action processing;
- Stage 3: transport those affine values through independent products and into one cross-prior block scale, allowing provisionally a bilinear interaction;
- Stage 4: identify reached-branch coefficients, prove their cocycle, and align the scales on nested support faces;
- Stage 5: compare product and sequential decompositions, eliminate the interaction, and invoke Faddeev's recursion to obtain Shannon entropy and hence mutual information.

The product-coefficient and branch tangent-space calculations are the two technical subarguments. A reader interested in the logical spine may use their lemma statements as interfaces. Product calibration is placed before branch calibration because this makes the scalar algebra easier to read. The early full-revelation normalisation and exact relabelling lemma below supply the coherence needed by that order of presentation.

A.1.1 Stage 1: posterior-law quotient and affine fibre values

For a finite action set A , let X_A be the set of finitely supported probability measures on $\Delta(A)$, and let

$$b(\mu) := \int_{\Delta(A)} s \, d\mu(s), \quad \mathcal{M}_q := \{\mu \in \mathsf{X}_A : b(\mu) = q\}.$$

Thus $\mathcal{M}_q$ is the mixture set of finite Bayes-plausible posterior laws with prior q , and $\mu_{q,P} \in \mathcal{M}_q$ for every finite channel P .

Lemma 20 (Coherent comparisons across blocks). *The relation on prior-channel pairs defined above is complete and transitive. It is unaffected by adding or deleting blocks other than the two being compared. In particular, for fixed q the induced relation on channels is a weak order, and*

$$q \succeq_P r \iff (q, P) \succeq (r, P).$$

For fixed alphabets, its graph is closed under entrywise convergence of the two channels and convergence of the two priors.

Proof. Completeness is Axiom (A1) in the two-block environment. To prove transitivity, suppose $(q_1, P_1) \succeq (q_2, P_2)$ and $(q_2, P_2) \succeq (q_3, P_3)$. Put the three pairs in the common environment $P_1 \sqcup P_2 \sqcup P_3$. Axiom (A3) transports the two hypotheses to that environment, Axiom (A1) composes them there, and Axiom (A3) deletes the middle block. Applying Axiom (A3) with the ordered pair of block labels reversed gives

$$(q_2, P_2) \succeq (q_1, P_1) \iff q_2^1 \succeq_{P_1 \sqcup P_2} q_1^0,$$

so reversal is also canonical. The displayed within-channel equivalence is the duplication clause of Axiom (A3). Finally, a block channel is a continuous affine function of its component channels. Closedness on fixed alphabets is therefore exactly Axiom (A2) applied to the common block environment. $\square$

Lemma 21 (Posterior-law quotient). *Fix a full-support $q \in \Delta(A)$. If finite channels $P : A \rightarrow \Delta(X)$ and $Q : A \rightarrow \Delta(Y)$ satisfy*

$$\mu_{q,P} = \mu_{q,Q},$$

then they are mutually obtainable by record post-processing. Consequently

$$(q, P) \sim (q, Q),$$

and either channel may be substituted for the other in every block comparison at prior q . Hence the fixed- q order of channels descends to a well-defined weak order $\succeq_q$ on $\mathcal{M}_q$.

Proof. We give the finite matching argument because it also handles repeated and zero-probability records. Put

$$m_P(x) := \sum_a q(a)P(x | a), \quad m_Q(y) := \sum_a q(a)Q(y | a).$$

For every posterior s occurring with positive probability, let

$$X_s := \{x : m_P(x) > 0, r_x^{q,P} = s\}, \quad Y_s := \{y : m_Q(y) > 0, r_y^{q,Q} = s\}.$$

Equality of posterior laws gives the common positive mass

$$\lambda_s := \sum_{x \in X_s} m_P(x) = \sum_{y \in Y_s} m_Q(y).$$

For $x \in X_s$, define a stochastic record kernel by

$$T(y | x) := \begin{cases} m_Q(y)/\lambda_s, & y \in Y_s, \\ 0, & y \notin Y_s. \end{cases}$$

Choose arbitrary stochastic rows at records x with $m_P(x) = 0$. Full support of q implies that such a record has $P(x | a) = 0$ for every a .

If $y \in Y_s$, Bayes' rule gives

$$\sum_{x \in X_s} P(x | a) = \frac{\lambda_s s(a)}{q(a)}, \quad Q(y | a) = \frac{m_Q(y)s(a)}{q(a)}.$$

It follows that

$$(PT)(y | a) = \sum_{x \in X_s} P(x | a) \frac{m_Q(y)}{\lambda_s} = Q(y | a).$$

Both sides vanish when $m_Q(y) = 0$, again by full support. Thus $PT = Q$. Interchanging P and Q constructs T' with $QT' = P$.

Axiom (A4) applied to T and T' yields the two weak comparisons; Lemma 20 puts the reverse comparison into the same block order and gives indifference. If P, P' have the same posterior law, place $(q, P), (q, P')$ and the two objects being compared in one finite block environment. Transitivity then permits replacement of P by P' on either side. The quotient order is consequently independent of the implementing channel and inherits completeness and transitivity from Lemma 20. $\square$

Every element of $\mathcal{M}_q$ is implementable when q has full support. Indeed, if

$$\mu = \sum_{k=1}^n \lambda_k \delta_{s_k}, \quad \lambda_k > 0, \quad \sum_k \lambda_k s_k = q,$$

then

$$C_\mu(k \mid a) := \frac{\lambda_k s_k(a)}{q(a)} \quad (8)$$

defines a channel and $\mu_{q, C_\mu} = \mu$.

For channels $P : A \rightarrow \Delta(X)$ and $R : A \rightarrow \Delta(Z)$ and $t \in [0, 1]$, let $tP \oplus (1-t)R$ be the publicly tagged mixture: it reports which component was selected as well as that component's record. Its posterior law is

$$\mu_{q, tP \oplus (1-t)R} = t\mu_{q, P} + (1-t)\mu_{q, R}. \quad (9)$$

Lemma 22 (Public-mixture independence). *Let q have full support. For all finite channels P, Q, R on the action set A and every $t \in (0, 1)$,*

$$(q, P) \succeq (q, Q) \iff (q, tP \oplus (1-t)R) \succeq (q, tQ \oplus (1-t)R).$$

Equivalently, the quotient order on $\mathcal{M}_q$ satisfies mixture independence.

Proof. If necessary, embed the record alphabets of P, Q, R into one tagged disjoint union and fill the unused coordinates with zeros. This does not change any posterior law, so Lemma 21 makes the padding neutral.

Let $C : A \rightarrow \Delta(\{0, 1\})$ be the uninformative first-stage channel with $C(0 \mid a) = t$ for every a . Both branches are reached, and their posterior is q . The compounds

$$C \triangleright \{P, R\}, \quad C \triangleright \{Q, R\}$$

have the posterior laws on the right-hand side of (9); they differ from the displayed public mixtures only by record labels. If $(q, P) \succeq (q, Q)$, Axiom (A6) applied to branch 0, together with reflexivity on the common branch 1, gives the aggregate comparison.

Conversely, suppose the aggregate comparison holds but $(q, P) \not\succeq (q, Q)$. Completeness gives $(q, Q) \succ (q, P)$. Apply the strict clause of Axiom (A6) with the two continuation profiles reversed. Branch 0 is reached with probability $t > 0$, so it yields the strict reverse aggregate comparison, contradicting the assumed weak one. Posterior-law substitution transfers both conclusions between the compounds and the tagged mixtures. $\square$

Lemma 23 (Affine posterior-law representation). *For every full-support $q \in \Delta(A)$ there is an affine functional*

$$F_q : \mathcal{M}_q \longrightarrow \mathbb{R}$$

such that, for all finite channels P, Q on A ,

$$(q, P) \succeq (q, Q) \iff F_q(\mu_{q, P}) \geq F_q(\mu_{q, Q}). \quad (10)$$

If $|A| \geq 2$, F_q is unique up to a positive affine transformation. If $|A| = 1$, $\mathcal{M}_q$ is a singleton and we take $F_q \equiv 0$.

Proof. By Lemma 21 and (8), $\succeq_q$ is a weak order on the mixture set $\mathcal{M}_q$. Lemma 22 gives mixture independence. It remains only to verify Herstein–Milnor's segment-continuity condition. This is where primitive Axiom (A2) is used; no topology on the collection of all finite posterior laws is needed.

Fix $\mu, \nu, \rho \in \mathcal{M}_q$ and choose finite implementing channels P, Q, R . For $t \in [0, 1]$ define

$$M_t := tP \oplus (1 - t)Q.$$

All M_t have the single fixed record alphabet given by the tagged union of the record alphabets of P and Q . If $t_n \rightarrow t$, then $M_{t_n} \rightarrow M_t$ entrywise. After adjoining the fixed block R , all comparisons occur in one fixed finite action and record alphabet. Axiom (A2) therefore makes both sets

$$\{t \in [0, 1] : M_t \succeq_q R\}, \quad \{t \in [0, 1] : R \succeq_q M_t\}$$

closed. By (9), these are exactly

$$\{t : t\mu + (1 - t)\nu \succeq_q \rho\}, \quad \{t : \rho \succeq_q t\mu + (1 - t)\nu\}.$$

Thus the Herstein–Milnor mixture-space theorem [30] applies and supplies an affine representation F_q . When $|A| \geq 2$, Axiom (A8) makes the order nonconstant, so the usual positive-affine uniqueness conclusion applies. For a singleton action set, Bayes plausibility forces every posterior law to equal δ_q . $\square$

The preceding proof uses only closedness along each particular public-mixture segment. It neither assumes nor proves closedness under arbitrary weakly convergent sequences whose support sizes grow. It also requires no pointwise integrand on $\Delta(A)$: every subsequent calculation uses only the affinity of F_q on $\mathcal{M}_q$ and its linear part on the affine hull.

Let

$$\delta_q \in \mathcal{M}_q$$

be the posterior law of the uninformative channel and let

$$\chi_q := \sum_{a \in A} q(a)\delta_{e_a} \in \mathcal{M}_q$$

be the posterior law of full revelation.

Lemma 24 (Canonical fibre normalization). *The representatives in Lemma 23 may be chosen so that*

$$F_q(\delta_q) = 0, \quad F_q(\mu) \geq 0 \quad (\mu \in \mathcal{M}_q), \quad (11)$$

and, whenever $|A| \geq 2$,

$$F_q(\chi_q) = 1. \quad (12)$$

These two anchors select a unique affine representative on every nondegenerate full-support fibre. On a singleton fibre we retain $F_q \equiv 0$.

Proof. Every channel P can be record-postprocessed to the uninformative channel by deleting its record. Axiom (A4) and (10) therefore imply

$$F_q(\mu_{q,P}) \geq F_q(\delta_q).$$

Every law in $\mathcal{M}_q$ is implemented by (8), so δ_q is a minimum of F_q on the whole fibre. Subtract $F_q(\delta_q)$ to obtain (11).

If $|A| \geq 2$, Axiom (A8) says that full revelation is strictly preferred to no information at every full-support prior. Hence $F_q(\chi_q) > 0$. Dividing by this positive number gives (12) without changing any comparison. If two affine representatives have both anchor values, positive-affine uniqueness forces the additive constant to be zero and the positive multiplier to be one. The singleton convention is immediate. $\square$

A.1.2 Stage 2: transport across priors and support faces

The remaining front-end issue is transport. The same argument handles renaming actions, deleting distinctions that the channel cannot detect, and passing to the positive support of a boundary prior. Keeping these facts in one lemma makes clear that all three are consequences of action processing, not extra invariance conventions.

For a bijection $\pi : A \rightarrow A'$, write $q\pi$ for the transported prior and $\pi_{\#}\mu$ for the posterior law obtained by transporting every atom of μ . If $B = \text{supp}(q)$, write q_B for the full-support restriction of q to B , P_B for the restricted channel, and $\pi_B\mu$ for the law obtained by restricting every posterior atom to B .

Lemma 25 (Relabelling, undetectable distinctions, and support transport). *The canonically normalized fibre values have the following properties.*

(a) Exact relabelling. *If q has full support and $\pi : A \rightarrow A'$ is a bijection, then*

$$F_{q\pi}^{A'}(\pi_{\#}\mu) = F_q^A(\mu) \quad (\mu \in \mathcal{M}_q). \quad (13)$$

(b) Undetectable distinctions. *Let $S : A \rightarrow B$ be an onto deterministic map, let q have full support, and suppose that $P : A \rightarrow \Delta(X)$ is constant on every fibre of S . Then*

$$(q, P) \sim (qS, S^q P). \quad (14)$$

In particular, if G_S reports only $S(a)$, then $(q, G_S) \sim (qS, \text{Id}_B)$.

(c) Support restriction. *For an arbitrary prior $q \in \Delta(A)$ with nonempty support $B := \text{supp}(q)$ and every channel P ,*

$$(q, P) \sim (q_B, P_B). \quad (15)$$

Consequently comparisons at q depend only on the rows indexed by B , and

$$(q, P) \succeq (q, Q) \iff (q_B, P_B) \succeq (q_B, Q_B). \quad (16)$$

Every atom of a law $\mu \in \mathcal{M}_q$ is supported on B , and the definition

$$F_q^A(\mu) := F_{q_B}^B(\pi_B\mu) \quad (17)$$

represents all continuation comparisons at the boundary prior q .

Proof. Part (a). Regard π as a deterministic action-report kernel. Axiom (A5) gives

$$(q, P) \succeq (q\pi, P\pi^{-1}),$$

where $(P\pi^{-1})(x \mid a') = P(x \mid \pi^{-1}(a'))$. Applying the same axiom to π^{-1} gives the reverse comparison, so the two pairs are indifferent. Apply this simultaneously to channels implementing arbitrary $\mu, \nu \in \mathcal{M}_q$ and use Lemma 20 in a common four-block environment. One obtains

$$F_q^A(\mu) \geq F_q^A(\nu) \iff F_{q\pi}^{A'}(\pi_{\#}\mu) \geq F_{q\pi}^{A'}(\pi_{\#}\nu).$$

Thus $G(\mu) := F_{q\pi}^{A'}(\pi_{\#}\mu)$ is an affine representation of the same nonconstant order as F_q^A . Relabelling sends δ_q to $\delta_{q\pi}$ and χ_q to $\chi_{q\pi}$. The two anchor equalities in Lemma 24, followed by affine uniqueness, give $G = F_q^A$. If A is a singleton, both sides vanish.

Part (b). Since q has full support and S is onto, qS has full support. One direction of (14) is Axiom (A5). For the reverse direction, define the Bayesian refinement kernel

$$R(a \mid b) := \begin{cases} q(a)/(qS)(b), & S(a) = b, \\ 0, & S(a) \neq b. \end{cases}$$

Then $(qS)R = q$. Since P is constant on each fibre of S , the pushforward S^qP has at b precisely the common row of P on $S^{-1}(b)$, and pushing it back through R recovers P . Axiom (A5) applied to (qS, S^qP) and R therefore gives the reverse weak comparison. For the last assertion, G_S is S -measurable and its pushforward is Id_B .

Part (c). Choose $b_0 \in B$ and let $S : A \rightarrow B$ fix every $b \in B$ and send every $a \notin B$ to b_0 . Since q assigns zero mass outside B ,

$$qS = q_B, \quad S^qP = P_B.$$

Axiom (A5) gives $(q, P) \succeq (q_B, P_B)$. Conversely, let $\iota : B \hookrightarrow A$ be the inclusion. Then $q_B\iota = q$. The exact Bayesian completion equation pins down the reported channel on B and leaves all zero-mass rows in $A \setminus B$ arbitrary. The all-completions clause of Axiom (A5) therefore permits us to choose the completion to be the original P , yielding $(q_B, P_B) \succeq (q, P)$. This proves (15); applying it to P and Q in a common block environment proves (16). It also shows immediately that channels agreeing on B are indifferent at q .

Finally, if $\mu = \sum_k \lambda_k \delta_{s_k} \in \mathcal{M}_q$ with $\lambda_k > 0$, then for $a \notin B$,

$$0 = q(a) = \sum_k \lambda_k s_k(a).$$

Nonnegativity forces $s_k(a) = 0$ for every k . Thus every atom lies on the face $\Delta(B)$ and $\pi_B\mu \in \mathcal{M}_{q_B}$. The comparison equivalence just proved and the full-support representation on B show that (17) represents precisely the boundary continuation order. This definition is also compatible with relabelling by part (a). $\square$

At this point every later calculation may work entirely with the affine functionals F_q , their linear parts on the affine hulls of $\mathcal{M}_q$, and the exact transport identities above. No global posterior-law continuity statement and no posterior-separable integrand are needed. From now on ambient action-set superscripts are suppressed unless two action sets occur in the same formula; a boundary-prior subscript always means the representative transported from its positive support. The next two stages derive product and branch scale coherence, with Axiom (A6) providing the essential cardinal alignment across reached branches.

A.1.3 Stage 3: products and a common comparison scale

The value F_q compares experiments only within the fibre of one prior. Embedding experiments from two priors into their product fibre will put both values on one cardinal comparison scale. We first derive the only separability fact about independent products needed for that construction. Write $P \succeq_q Q$ for $(q, P) \succeq (q, Q)$, and similarly for strict preference. For a channel $P : A \rightarrow \Delta(X)$, let $\tilde{P}$ denote its lift to $A \times B$, constant in the B -coordinate; for $R : B \rightarrow \Delta(Y)$, let $\tilde{R}$ denote the symmetric lift, constant in the A -coordinate. Posterior laws may be substituted throughout by Lemma 21; in particular, adding null records and permuting record labels has no effect on a comparison.

Lemma 26 (Cancellation of a common independent background). *Let $p \in \Delta(A)$ and $r \in \Delta(B)$ have full support. For channels P, Q on A and a channel R on B ,*

$$P \otimes R \succeq_{p \otimes r} Q \otimes R \iff P \succeq_p Q. \quad (\text{BI})$$

The symmetric assertion holds with the two coordinates interchanged. The equivalence, including its strict form, remains valid when a posterior reached in the background coordinate lies on a proper support face.

Proof. Pad P and Q by null records so that they have the same record alphabet. Regard the lift $\tilde{R}$ of R as a first-stage experiment on $A \times B$, followed on every reached branch by $\tilde{P}$, respectively $\tilde{Q}$. Up to the harmless permutation of the two record coordinates, the resulting compounds are $P \otimes R$ and $Q \otimes R$.

If y is reached under R , its posterior on $A \times B$ is

$$p \otimes r_y, \quad r_y := r_y^{r,R}.$$

The coordinate projection $A \times \text{supp}(r_y) \rightarrow A$ and the support-transport clauses of Lemmas 25 and 25 give

$$(p \otimes r_y, \tilde{P}) \sim (p, P), \quad (p \otimes r_y, \tilde{Q}) \sim (p, Q). \quad (\text{BI}')$$

Thus every reached branch has the same weak comparison as P versus Q at p . If $P \succeq_p Q$, the weak clause of Axiom (A6) applied to these branches proves the forward implication in (BI).

For the converse, suppose that $P \not\succeq_p Q$. Completeness gives $Q \succ_p P$. By (BI'), the reversed comparison is strict on every reached branch; at least one such branch has positive mass. The strict clause of Axiom (A6) therefore gives $Q \otimes R \succ_{p \otimes r} P \otimes R$, contradicting $P \otimes R \succeq_{p \otimes r} Q \otimes R$. This proves the converse and also records why weak branch monotonicity alone would not suffice. Interchange the coordinates for the symmetric assertion. $\square$

For finite posterior laws define

$$\left(\sum_i m_i \delta_{u_i} \right) \otimes \left(\sum_j n_j \delta_{v_j} \right) := \sum_{i,j} m_i n_j \delta_{u_i \otimes v_j}.$$

Bayes' rule gives

$$\mu_{p \otimes r, P \otimes R} = \mu_{p,P} \otimes \mu_{r,R}. \quad (\text{PL})$$

Lemma 27 (Product gauge). *There are positive rescalings $G_q = c_q F_q$ of the zero-normalised affine representatives and a single number $\kappa \in \mathbb{R}$ such that*

$$G_{p \otimes r}(\mu \otimes \nu) = G_p(\mu) + G_r(\nu) + \kappa G_p(\mu) G_r(\nu) \quad (\text{QA})$$

for all full-support priors p, r and all $\mu \in \mathcal{M}_p$, $\nu \in \mathcal{M}_r$. These rescalings can be chosen exactly invariant under bijections. If a factor is a singleton, its representative is zero and the same formula holds. Finally,

$$1 + \kappa G_r(\nu) > 0 \quad (\text{POS})$$

whenever the r -fibre is nonconstant.

Proof. We give the algebra because both the common value of κ and the strict inequality (POS) are used later. Initially retain the notation F_q . For nondegenerate p, r , put

$$L_{p,r}(\mu, \nu) := F_{p \otimes r}(\mu \otimes \nu), \quad x := F_p(\mu), \quad y := F_r(\nu).$$

The map $L_{p,r}$ is separately affine. By Lemma 26, every first-coordinate slice represents the F_p -order and every second-coordinate slice represents the F_r -order. Affine-utility uniqueness, first on a slice and then as the other coordinate varies affinely, therefore gives unique constants $A_{p,r}, B_{p,r} > 0$ and $C_{p,r} \in \mathbb{R}$ such that

$$L_{p,r}(\mu, \nu) = A_{p,r}x + B_{p,r}y + C_{p,r}xy. \quad (18)$$

Here is the coefficient argument. Fixing ν first gives $L_{p,r} = \alpha(\nu)x + \gamma(\nu)$ with $\alpha(\nu) > 0$. At $\mu = \delta_p$, the function $\gamma(\nu) = L_{p,r}(\delta_p, \nu)$ is an affine representative of the F_r -order and vanishes at

δ_r ; hence $\gamma(\nu) = B_{p,r}y$ for some $B_{p,r} > 0$. Choose $\bar{\mu}$ with $\bar{x} := F_p(\bar{\mu}) \neq 0$. The other slice has the form

$$L_{p,r}(\bar{\mu}, \nu) = A_{p,r}\bar{x} + D_{p,r}y,$$

where $A_{p,r} > 0$ is its value slope at $\nu = \delta_r$. Comparing this with $\alpha(\nu)\bar{x} + B_{p,r}y$ shows that $\alpha(\nu) = A_{p,r} + C_{p,r}y$, where $C_{p,r} := (D_{p,r} - B_{p,r})/\bar{x}$. This proves (18); the symmetric calculation shows that the two slice slopes obey

$$A_{p,r} + C_{p,r}y > 0, \quad B_{p,r} + C_{p,r}x > 0 \quad (19)$$

throughout the two value ranges.

Exact relabelling makes the evaluations under the two associations of a triple product equal. Comparing the coefficients of x, y, z gives

$$A_{p \otimes r, s} A_{p, r} = A_{p, r \otimes s}, \quad (20)$$

$$A_{p \otimes r, s} B_{p, r} = B_{p, r \otimes s} A_{r, s}, \quad (21)$$

$$B_{p \otimes r, s} = B_{p, r \otimes s} B_{r, s}. \quad (22)$$

The coordinate swap similarly gives

$$B_{r, p} = A_{p, r}, \quad A_{r, p} = B_{p, r}, \quad C_{r, p} = C_{p, r}. \quad (\text{SW})$$

This coefficient comparison is legitimate: the laws $(1-t)\delta_p + t\chi_p$ make x range over a nondegenerate interval, and the same holds for y and z . Let $\rho(p, r) := B_{p,r}/A_{p,r}$. Dividing (21) by (20) yields

$$\rho(p, r) = \rho(p, r \otimes s) A_{r, s}.$$

Fix a nondegenerate reference prior q_0 and set $g(r) := \rho(q_0, r)$. Then

$$A_{r, s} = \frac{g(r)}{g(r \otimes s)}. \quad (23)$$

The uniqueness of the coefficients in (18), applied after product relabellings, shows that g is invariant under bijections. Consequently the positive rescaling $G_p := g(p)F_p$ preserves exact relabelling. Equation (23) makes the new first coefficient equal to one, and (SW) makes the second coefficient equal to one as well. We henceforth use G for these gauged representatives.

Write the remaining coefficient as $\kappa_{p,r}$. Associativity of

$$x \oplus_{p,r} y := x + y + \kappa_{p,r}xy$$

and comparison of the coefficients of xy, xz, yz give

$$\kappa_{p,r} = \kappa_{p, r \otimes s}, \quad \kappa_{p \otimes r, s} = \kappa_{p, r \otimes s}, \quad \kappa_{p \otimes r, s} = \kappa_{r, s}. \quad (\text{KA})$$

(The xyz coefficient gives the product of the corresponding two sides and adds no restriction.) Taking $s = q_0$ in (KA) shows that $\kappa_{p,r} = \kappa_{r, q_0}$, so it is independent of p ; the swap in (SW) makes it symmetric, hence independent of r . Call the common value κ . After the gauge, (19) is exactly $1 + \kappa G_r(\nu) > 0$ and its symmetric counterpart.

For a singleton prior set $G \equiv 0$; the canonical bijections $A \cong A \times \{*\}$ give (QA). For a boundary prior, define G_r by exact transport from its positive-probability support, as in Lemma 25. This convention is relabelling invariant. $\square$

Corollary 28 (Exact relabelling after the product gauge). *For every bijection $\pi : A \rightarrow A'$, prior $q \in \Delta(A)$, and $\mu \in \mathcal{M}_q$,*

$$G_{q\pi}^{A'}(\pi_{\#}\mu) = G_q^A(\mu).$$

Proof. The initial equality is Lemma 25; the proof of Lemma 27 shows that its gauge factor is invariant under bijections. $\square$

Lemma 29 (Cross-prior block representation). *For arbitrary finite priors $q \in \Delta(A)$ and $r \in \Delta(B)$ and channels P, Q on the respective action sets,*

$$q^0 \succeq_{P \sqcup Q} r^1 \iff G_q(\mu_{q,P}) \geq G_r(\mu_{r,Q}). \quad (\text{BC})$$

Proof. First suppose that q, r have full support. By (QA) and $G_q(\delta_q) = G_r(\delta_r) = 0$,

$$G_{q \otimes r}(\mu_{q,P} \otimes \delta_r) = G_q(\mu_{q,P}), \quad G_{q \otimes r}(\delta_q \otimes \mu_{r,Q}) = G_r(\mu_{r,Q}).$$

The two laws on the left lie in the same fibre and are implemented by $P \otimes U_B$ and $U_A \otimes Q$, so the affine fibre representation compares them by the displayed numbers.

It remains only to remove the uninformative product coordinates. Projection $A \times B \rightarrow A$ and Axiom (A5) give $(q \otimes r, P \otimes U_B) \succeq (q, P)$. Conversely the stochastic embedding

$$T((a, b) \mid a') = \mathbf{1}_{\{a=a'\}} r(b)$$

sends q to $q \otimes r$ and its Bayesian pushforward sends P to $P \otimes U_B$; Axiom (A5) gives the reverse comparison. Thus the two pairs are indifferent. The same argument identifies $(q \otimes r, U_A \otimes Q)$ with (r, Q) . Block coherence and transitivity now prove (BC). Arbitrary priors follow by restricting both pairs and both representatives to their supports; singleton supports use the zero convention. The singleton record tags created by these projections and embeddings are the canonical relabellings covered by the convention at the start of the appendix. $\square$

A.1.4 Stage 4: sequential decomposition and coherent face scales

Changing a continuation on one reached branch changes the posterior law by a signed measure with zero total mass and zero barycentre. The following space is exactly the space of such feasible directions. For a nonempty finite set A , let

$$W_A := \left\{ \eta : \begin{array}{l} \eta \text{ is a finitely supported signed measure on } \Delta(A), \\ \eta(\Delta(A)) = 0, \quad \int p \, d\eta(p) = 0 \end{array} \right\}.$$

If $\iota : B \hookrightarrow A$ is an injection, let $i_\iota : W_B \rightarrow W_A$ push every posterior forward along ι ; for a literal subset $B \subseteq A$, write i_B for the inclusion case. Let L_q^A be the linear part of the affine functional G_q on the affine hull of $\mathcal{M}_q$.

Lemma 30 (Branch aggregation). *Let q have full support on A . For every reached posterior r with $B := \text{supp}(r)$ and $|B| \geq 2$, there is a unique $\beta_A(q, r) > 0$ satisfying*

$$L_q^A \circ i_B = \beta_A(q, r) L_{r_B}^B \quad \text{on } W_B. \quad (24)$$

It depends only on q, r and the face inclusion, not on the experiment that reaches r . If a first-stage channel P_1 has reached branches of masses m_x and posteriors r_x , then every continuation profile satisfies

$$G_q(\mu_{q, P_1 \triangleright \{Q^x\}}) = G_q(\mu_{q, P_1}) + \sum_{x: m_x > 0} m_x \beta_A(q, r_x) G_{r_x}(\mu_{r_x, Q^x}). \quad (\text{BA})$$

For singleton r_x , the continuation value is zero and the corresponding coefficient may be fixed by any positive convention.

Proof. We first record the common tangent space. If q has full support, then

$$\text{span}\{\mu - \nu : \mu, \nu \in \mathcal{M}_q\} = W_A. \quad (25)$$

Only the reverse inclusion needs proof. For $0 \neq \eta \in W_A$, write its Jordan decomposition as $\eta = \eta^+ - \eta^-$, where both positive parts have the same mass $c > 0$ and, after division by c , the same barycentre b . Choose $\varepsilon > 0$ small enough that $\tau = (q - \varepsilon b)/(1 - \varepsilon)$ belongs to $\Delta(A)$. Then

$$\mu^\pm := \frac{\varepsilon}{c}\eta^\pm + (1 - \varepsilon)\delta_\tau \in \mathcal{M}_q, \quad \eta = \frac{c}{\varepsilon}(\mu^+ - \mu^-),$$

which proves (25). The same argument intrinsically on a face gives the space W_B .

Fix a nondegenerate r supported on B . It is reachable from q : choose

$$0 < \varepsilon < \min \left\{ 1, \min_{b \in B} \frac{q(b)}{r(b)} \right\}$$

and use a two-record experiment whose first record has conditional probability $\varepsilon r(a)/q(a)$. That record has mass ε and posterior r .

Implement $\nu, \nu' \in \mathcal{M}_{r_B}$ by channels on B , extend their off-support rows arbitrarily to A , and pad their record alphabets if necessary. Vary only the continuation on the branch reaching r from ν to ν' . The aggregate posterior law changes by $\varepsilon i_B(\nu - \nu')$. Axiom (A6), support transport, and the strict clause give all three signs:

$$\begin{aligned} L_{r_B}^B(\nu - \nu') > 0 &\implies L_q^A i_B(\nu - \nu') > 0, \\ L_{r_B}^B(\nu - \nu') < 0 &\implies L_q^A i_B(\nu - \nu') < 0, \\ L_{r_B}^B(\nu - \nu') = 0 &\implies L_q^A i_B(\nu - \nu') = 0. \end{aligned}$$

The second line is the first applied after swapping the continuations. For the third, indifference supplies both weak branch comparisons, and the weak part of Axiom (A6) supplies both aggregate inequalities. By (25), the sign agreement extends by positive scaling to all of W_B . Positive trace orientation makes

$$L_{r_B}^B(\chi_{r_B} - \delta_{r_B}) = G_{r_B}(\chi_{r_B}) > 0,$$

so $L_{r_B}^B \neq 0$. Two nonzero linear functionals with the same positive half-space have the same kernel and are positive scalar multiples; this gives the unique coefficient in (24). Since both linear functionals in that equation are fixed before a reaching experiment is chosen, the coefficient is path-independent.

For the aggregation formula, put $B_x := \text{supp}(r_x)$ and

$$\nu_x := \pi_{B_x} \mu_{r_x, Q^x} \in \mathcal{M}_{(r_x)_{B_x}}.$$

The compound law differs from the first-stage law by

$$\mu_{q, P_1 \triangleright \{Q^x\}} - \mu_{q, P_1} = \sum_{x: m_x > 0} m_x i_{B_x}(\nu_x - \delta_{(r_x)_{B_x}}).$$

Apply L_q^A , use (24), and use $G_{r_x}(\delta_{r_x}) = 0$. Singleton summands vanish. This is (BA). $\square$

For an injection $\iota : B \hookrightarrow A$ and full-support $r \in \Delta(B)$, define

$$\beta_\iota(q, r) := \beta_A(q, \iota_{\#} r).$$

The image posterior $\iota_{\#} r$ is reachable by the two-record construction in the proof, and its support inclusion is precisely i_ι . Thus

$$L_q^A i_\iota = \beta_\iota(q, r) L_r^B \quad \text{on } W_B.$$

Lemma 31 (Cocycle, face coherence, and the chain gauge). *The coefficients of Lemma 30 admit positive scales a_q^A which are invariant under relabelling and satisfy*

$$\beta_\iota(q, r) = \frac{a_q^A}{a_r^B} \quad (\text{FS})$$

whenever $\iota : B \hookrightarrow A$, both sets are nondegenerate, q and r have full support on their respective sets, and $\beta_\iota(q, r)$ is the coefficient for reaching the face $\iota(B)$. Hence the representatives

$$V_q := \frac{G_q}{a_q}$$

satisfy the exact chain rule

$$V_q(\mu_{q, P_1 \triangleright \{Q^x\}}) = V_q(\mu_{q, P_1}) + \sum_{x: m_x > 0} m_x V_{r_x}(\mu_{r_x, Q^x}). \quad (\text{CR})$$

Singleton continuation terms are zero.

Proof. There are two normalization tasks: first solve the branch cocycle within each fixed action-set dimension, then remove the residual scalar attached to embedding an m -point face in an n -point simplex.

We first prove the cocycle before choosing scales. For nested injections $C \xrightarrow{j} B \xrightarrow{\iota} A$ and full-support priors q, r, s on A, B, C , respectively, the defining identities on W_C are

$$L_q^A i_{\iota \circ j} = \beta_\iota(q, r) L_r^B i_j = \beta_\iota(q, r) \beta_j(r, s) L_s^C.$$

The direct coefficient from q to s is unique because $L_s^C \neq 0$ by Axiom (A8). Therefore

$$\beta_{\iota \circ j}(q, s) = \beta_\iota(q, r) \beta_j(r, s). \quad (26)$$

This single identity covers full-support paths, full-to-face paths, and nested support faces.

For a fixed nondegenerate A , apply (26) to identity embeddings. Fix a full-support basepoint q_A and put

$$a_q^A := \beta_{\text{id}_A}(q, q_A).$$

Then

$$\beta_{\text{id}_A}(q, r) = \frac{a_q^A}{a_r^A}. \quad (\text{WS})$$

Choosing q_A to be uniform and using exact relabelling of G makes these initial scales invariant under bijections. They remain free up to one positive multiplier for each cardinality.

For an injection $\iota : B \hookrightarrow A$, define its residual defect by

$$K(\iota) := \beta_\iota(q, r) \frac{a_r^B}{a_q^A}.$$

Equation (WS) shows that this number is independent of q and r : changing either prior merely inserts and cancels the corresponding within-set coefficient. Equation (26) gives

$$K(\iota \circ j) = K(\iota) K(j).$$

Exact relabelling implies that $K(\iota)$ depends only on $n = |A|$ and $m = |B|$; write it $K_{n,m}$. Thus

$$K_{n,\ell} = K_{n,m} K_{m,\ell}, \quad K_{n,n} = 1.$$

Set $t_n := K_{n,2}$ and $t_2 := 1$. Then $K_{n,m} = t_n/t_m$. Replacing every a_q^A on an n -point set by $t_n a_q^A$ changes the defect to $K_{n,m} t_m/t_n = 1$, while preserving (WS) and relabel invariance. This proves (FS).

For a boundary prior, a_r means the scale on its support; the preceding construction says precisely that this agrees with the scale obtained by reaching that support as a face. Divide (BA) by a_q and use (FS). Every nondegenerate branch coefficient becomes

$$\frac{1}{a_q} \beta_A(q, r_x) G_{r_x} = \frac{1}{a_{r_x}} G_{r_x} = V_{r_x}.$$

Singleton values vanish, so their scales and coefficients may be fixed arbitrarily. This proves (CR). $\square$

A.1.5 Stage 5: compatibility and entropy identification

The product and sequential evaluations of full revelation now meet. Their compatibility first removes the provisional product interaction and then leaves a strongly additive uncertainty functional.

Lemma 32 (Interaction collapse and universal chain scale). *The coefficient κ in (QA) is zero. Moreover there is a single $C > 0$ such that $a_q = C$ for every nondegenerate finite prior, read on its support. The same value may be assigned to singleton priors. Consequently the final representatives $V_q = G_q/C$ satisfy both*

$$V_{p \otimes r}(\mu \otimes \nu) = V_p(\mu) + V_r(\nu), \quad (\text{PA})$$

$$V_q(\mu_{q, P_1 \triangleright \{Q^x\}}) = V_q(\mu_{q, P_1}) + \sum_{x: m_x > 0} m_x V_{r_x}(\mu_{r_x, Q^x}). \quad (\text{CR}')$$

The cross-prior bridge of Lemma 29 is valid with V in place of G .

Proof. In this proof only, put

$$H_G(q) := G_q(\chi_q), \quad Z(q) := 1 + \kappa H_G(q).$$

For nondegenerate q , Axiom (A8) and zero normalisation give $H_G(q) > 0$, while the strict slice-slope (POS) gives

$$Z(q) > 0. \quad (27)$$

For a singleton, $H_G = 0$ and $Z = 1$.

Product revelation versus sequential revelation. For nondegenerate full-support p, r , product quasi-additivity gives

$$H_G(p \otimes r) = H_G(p) + H_G(r) + \kappa H_G(p) H_G(r). \quad (\text{QH})$$

Compute the same full revelation by first revealing the A -coordinate and then the B -coordinate. The first-stage law is $\chi_p \otimes \delta_r$ and has value $H_G(p)$ by (QA). After action a is revealed, the posterior is $e_a \otimes r$ on the face $\{a\} \times B$; support restriction and relabelling identify the continuation value with $H_G(r)$. Every such branch has coefficient $a_{p \otimes r}/a_r$ by (FS). Since the branch weights sum to one, Lemma 30 gives

$$H_G(p \otimes r) = H_G(p) + \frac{a_{p \otimes r}}{a_r} H_G(r). \quad (\text{SA})$$

Comparing (QH) and (SA), and using $H_G(r) > 0$, yields

$$\frac{a_{p \otimes r}}{a_r} = Z(p). \quad (\text{R1})$$

Revealing B first gives symmetrically

$$\frac{a_{p \otimes r}}{a_p} = Z(r). \quad (\text{R2})$$

Thus $a_r Z(p) = a_p Z(r)$ for every pair of nondegenerate priors, even on different finite sets. By (27), there is a universal $C > 0$ such that

$$a_q = CZ(q). \quad (\text{AS})$$

Assign $a = C$ on singleton supports.

If $\kappa = 0$, then $Z \equiv 1$ and the conclusion already follows. It remains to rule out $\kappa \neq 0$.

The grouping weight. Let q have full support and let $\mathcal{P} = \{A_1, \dots, A_m\}$ partition its action set into nonempty cells. Put

$$p_i = q(A_i), \quad q^i = q(\cdot \mid A_i), \quad p = (p_1, \dots, p_m).$$

If $C_{\mathcal{P}}$ reports only the cell, undetectable-distinctions neutrality gives

$$(q, C_{\mathcal{P}}) \sim (p, \text{Id}_{\{1, \dots, m\}}).$$

The cross-prior bridge therefore identifies the coarse value as

$$G_q(\mu_{q, C_{\mathcal{P}}}) = H_G(p). \quad (\text{BG})$$

Append full revelation within each cell. The final law is χ_q , and the branch formula together with (AS) gives

$$H_G(q) = H_G(p) + Z(q) \sum_i p_i \frac{H_G(q^i)}{Z(q^i)}. \quad (\text{GR})$$

Here singleton cells contribute zero.

Set $w(q) := 1/Z(q) > 0$. Because $\kappa \neq 0$ in the present paragraph, $\kappa H_G(x) = Z(x) - 1$. Substitution into (GR) gives

$$Z(q) - 1 = Z(p) - 1 + Z(q) \sum_i p_i \left(1 - \frac{1}{Z(q^i)}\right).$$

Cancelling and inverting yields the positive grouping-weight equation

$$w(q) = w(p) \sum_i p_i w(q^i). \quad (\text{W})$$

All distributions in what follows are read on their positive-probability supports. For $u \otimes v$, partition by the first coordinate: the coarse distribution is u and every conditional distribution is v . Thus (W) gives

$$w(u \otimes v) = w(u)w(v). \quad (\text{M})$$

Two groupings force the weight to be constant. Take arbitrary finite probability vectors u, v and write $x = w(u)$ and $y = w(v)$. Let T be the law of a triple (ℓ, z_1, z_2) : ℓ is uniform on $\{0, 1\}$, and conditional on $\ell = 0$ the pair (z_1, z_2) has law $u \otimes u$, while conditional on $\ell = 1$ it has law $v \otimes v$. In tagged notation,

$$T := \frac{1}{2}(u \otimes u)^0 \sqcup \frac{1}{2}(v \otimes v)^1.$$

Grouping first by ℓ and using (M) gives

$$w(T) = w(\frac{1}{2}, \frac{1}{2}) \frac{x^2 + y^2}{2}. \quad (\text{E1})$$

Instead group first by (ℓ, z_1) . That marginal is $S = \frac{1}{2}u^0 \sqcup \frac{1}{2}v^1$, so

$$w(S) = w(\frac{1}{2}, \frac{1}{2}) \frac{x + y}{2}.$$

The conditional law of z_2 is u when $\ell = 0$ and v when $\ell = 1$. A second use of (W) therefore gives

$$w(T) = w(S) \frac{x+y}{2} = w(\tfrac{1}{2}, \tfrac{1}{2}) \left(\frac{x+y}{2} \right)^2. \quad (\text{E2})$$

The common leading factor is positive. Comparing (E1) and (E2) yields $(x-y)^2 = 0$. Thus w is constant on all finite probability vectors. Its singleton value is one, so $w \equiv 1$.

It follows that $Z \equiv 1$, hence $\kappa H_G(q) = 0$ for every q . Axiom (A8) supplies a nondegenerate q with $H_G(q) > 0$, so $\kappa = 0$, contradicting the temporary alternative. Equation (AS) now gives $a_q = C$ universally. Dividing the product and branch formulas by C proves (PA) and (CR'). Since the division is by one common positive number, it also preserves the cross-prior bridge. $\square$

Entropy identification.

Lemma 33 (Entropy reduction and Faddeev identification). *Define the value of full revelation by*

$$\mathcal{H}(q) := V_q(\chi_q),$$

so $\mathcal{H}(q) = H_G(q)/C$, with boundary priors read on their support. Then every finite channel $P : A \rightarrow \Delta(X)$ satisfies

$$V_q(\mu_{q,P}) = \mathcal{H}(q) - \sum_{x:m_{q,P}(x)>0} m_{q,P}(x) \mathcal{H}(r_x^{q,P}). \quad (\text{ER})$$

Moreover,

$$\mathcal{H}(q) = \alpha H(q)$$

for a single $\alpha > 0$. Consequently

$$V_q(\mu_{q,P}) = \alpha I_{q,P}(A; X). \quad (\text{MI})$$

Proof. Entropy reduction. Assume first that q has full support. Append full revelation after P . The final posterior law is χ_q , while the continuation after record x has full-revelation law χ_{r_x} on the support of r_x . The exact chain rule (CR') therefore gives

$$\mathcal{H}(q) = V_q(\mu_{q,P}) + \sum_{x:m_{q,P}(x)>0} m_{q,P}(x) \mathcal{H}(r_x),$$

which is (ER). A boundary prior is reduced to its support; null action rows and null records affect neither side.

Strong additivity. Let $\mathcal{P} = \{A_1, \dots, A_m\}$ be a partition of the support of q , and put

$$p_i = q(A_i), \quad q^i = q(\cdot \mid A_i), \quad p = (p_1, \dots, p_m).$$

The channel $C_{\mathcal{P}}$ which reports the cell is neutral to the corresponding full revelation on the cell index:

$$(q, C_{\mathcal{P}}) \sim (p, \text{Id}_{\{1, \dots, m\}})$$

by Lemma 25. The common-scale cross-prior bridge from Lemma 32 therefore gives

$$V_q(\mu_{q,C_{\mathcal{P}}}) = \mathcal{H}(p).$$

Applying (ER) to $C_{\mathcal{P}}$ yields

$$\mathcal{H}(q) = \mathcal{H}(p) + \sum_i p_i \mathcal{H}(q^i). \quad (\text{FA})$$

Thus $\mathcal{H}$ is strongly additive. Relabelling invariance of V makes $\mathcal{H}$ symmetric; support restriction makes it expansive under adding zero-probability states; and $\mathcal{H}(\delta_a) = 0$. Record processing

makes every experiment weakly preferable to its no-information garbling, so $V_q(\mu) \geq 0$ on every fibre and hence $\mathcal{H}(q) \geq 0$. Axiom (A8) gives strict positivity at every nondegenerate full-support prior. Although (FA) was derived by partitioning the positive support, it also holds with zero outer weights: delete the null cells by support restriction, apply the displayed recursion, and reinsert them by expansibility.

Continuity: the binary case. Write $h(t) := \mathcal{H}(t, 1 - t)$. We derive, rather than assume, its continuity from Axiom (A2). Fix $c \geq 0$. Choose a full-support binary prior θ with $\mathcal{H}(\theta) > 0$, as supplied by Axiom (A8). Choose $n \geq 1$ and $\lambda \in [0, 1]$ so that

$$c = \lambda n \mathcal{H}(\theta).$$

For $\theta_n := \theta^{\otimes n}$, product additivity (PA) gives $\mathcal{H}(\theta_n) = n \mathcal{H}(\theta)$. Let $E_{n,\lambda}$ reveal the n -bit action fully with probability λ and report a common erasure symbol otherwise. Its posterior law is $\lambda \chi_{\theta_n} + (1 - \lambda) \delta_{\theta_n}$, so affinity gives

$$V_{\theta_n}(\mu_{\theta_n, E_{n,\lambda}}) = c.$$

Now fix the block environment

$$K_c := \text{Id}_{\{0,1\}} \sqcup E_{n,\lambda}.$$

Let q_t^0 put the binary prior $(t, 1 - t)$ on its first block and let θ_n^1 put θ_n on its second block. The cross-prior bridge says

$$\begin{aligned} h(t) \geq c &\iff q_t^0 \succeq_{K_c} \theta_n^1, \\ h(t) \leq c &\iff \theta_n^1 \succeq_{K_c} q_t^0. \end{aligned}$$

The environment is fixed and $t \mapsto q_t^0$ is continuous, so Axiom (A2) makes both level sets closed. When $c < 0$, the superlevel set is all of $[0, 1]$ and the sublevel set is empty because $h \geq 0$. Thus every upper and lower level set of h is closed, and h is continuous.

Continuity on every finite simplex. Proceed by induction on the number of coordinates. For $q = (q_1, (1 - q_1)\bar{q})$ with $q_1 < 1$, recursion (FA) for the partition $\{\{1\}, \{2, \dots, n\}\}$ gives

$$\mathcal{H}(q) = h(q_1) + (1 - q_1) \mathcal{H}(\bar{q}). \quad (28)$$

The right side is continuous away from $q_1 = 1$ by binary continuity and the induction hypothesis. At $q_1 = 0$, support restriction and $h(0) = 0$ give the same formula on the face. At $q_1 = 1$, the induction hypothesis makes $\mathcal{H}$ bounded on the compact $(n - 1)$ -simplex, so the second term in (28) tends to zero and the first tends to $h(1) = 0$. Relabelling handles the other faces. Hence $\mathcal{H}$ is continuous on each closed finite simplex.

We have now verified all hypotheses of the strong-additivity form of Faddeev's theorem: continuity on each finite simplex, invariance under bijections, expansibility, vanishing on point masses, and (FA). The finite entropy characterisation [8, Theorem 6] therefore gives

$$\mathcal{H}(q) = \alpha H(q)$$

for some $\alpha \geq 0$. Axiom (A8) forces $\alpha > 0$. Substitution in (ER) gives

$$V_q(\mu_{q,P}) = \alpha \left(H(q) - \sum_x m_{q,P}(x) H(r_x^{q,P}) \right) = \alpha I_{q,P}(A; X),$$

where the sum is over positive-mass records. This proves (MI). $\square$

Completion of Proposition 19. After Lemma 32, the cross-prior bridge is represented by V . Lemma 33 identifies this value as αI with one $\alpha > 0$. Hence, for arbitrary pairs (q, P) and (r, Q) ,

$$q^0 \succeq_{P \sqcup Q} r^1 \iff I_{q,P} \geq I_{r,Q}, \quad (\text{BMI})$$

including boundary priors by support restriction. For one fixed channel, the duplication clause of Axiom (A3) identifies $q \succeq_P q'$ with the left side of (BMI) for $Q = P$, proving the first assertion. For the block clause, Axiom (A3) deletes all irrelevant blocks and (BMI) compares the two remaining pairs. A lottery supported on block i has a constant block tag, so

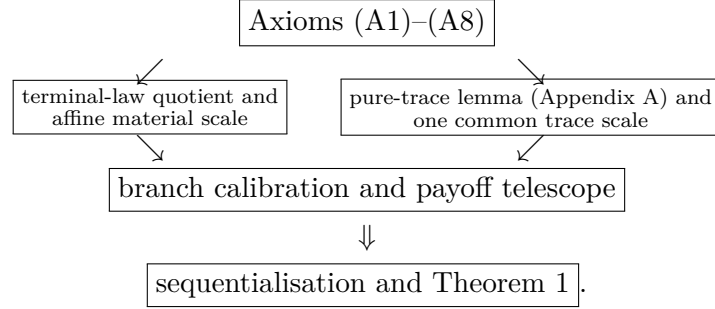
$$I_{q_i^i, \bigsqcup_{k \in J} P_k} = I_{q_i, P_i},$$

and similarly for block j . This is exactly the stated common-scale block comparison. $\square$

B Completion of the representation theorem

Appendix A proves the one substantial auxiliary input: the pure-trace lemma on the constant-payoff domain. This appendix combines that result with the affine material scale, calibrates deterministic payoff branches, and completes Theorem 1. We also use the ordinary mixture-space theorem: a complete and transitive mixture order satisfying independence and segment calibration has an affine representation.

The logical order is therefore



The left branch identifies the value of terminal payoffs and fixes a common material scale. The right branch is proved once on the constant-payoff domain. They meet in the deterministic branch calculation, which aligns material increments with the trace scale and telescopes across all reached branches.

B.1 Preliminaries

We first dispose of three pieces of bookkeeping. They will be used without further comment.

Recall that $(q, K) \succeq (p, L)$ denotes the paper's canonical two-block comparison. For channels E, G at a common prior q , write $E \succeq_q G$ for $(q, E) \succeq (q, G)$. A *finite experiment at q* is a finite payoff-record channel paired with that prior; its marked terminal law is denoted by η_E .

Lemma 34 (Blocks, supports, and dummies). *Suppose Axioms (A1)–(A5) hold.*

- (a) *After identifying alternatives related by invertible action or record relabellings, the derived comparison of pairs is complete and transitive. If either member of a pair comparison is replaced by an indifferent alternative, the comparison is unchanged.*
- (b) *If $S = \text{supp } q$, then (q, K) is indifferent to the restriction of (q, K) to S .*
- (c) *If $s \in \Delta(B)$ and $K^\uparrow(o, r \mid a, b) := K(o, r \mid a)$, then*

$$(q \otimes s, K^\uparrow) \sim (q, K). \quad (29)$$

Proof. Place any finite collection of alternatives in one finite block environment. Axiom (A3) removes unused blocks, and Axiom (A1) supplies completeness and transitivity there. Invertible action and record relabellings are neutral by applying Axioms (A4) and (A5) in both directions. In particular, swapping both tags identifies the two orientations of a two-block comparison. This proves part (a); a three-block environment gives transitivity and a four-block environment gives replacement.

For part (b), report an action in S by its inclusion in A . In the other direction, report $a \in S$ when the realised action is a , and choose an arbitrary element of S off the support. Both reports satisfy (2); rows attached to null reported actions may be completed arbitrarily. Axiom (A5) gives the two comparisons.

For part (c), first project (a, b) to a . Conversely, given a , draw b independently from s and report (a, b) . Again the two joint-law identities in (2) are exact, so Axiom (A5) applies in both directions. Part (a) completes both arguments. $\square$

Fix for the moment a full-support prior $q \in \Delta(A)$. For a joint channel K , write

$$p_K(o, r) := \sum_a q(a) K(o, r | a), \quad \pi_{or}^K(a) := \frac{q(a) K(o, r | a)}{p_K(o, r)}$$

when $p_K(o, r) > 0$, and define its *marked terminal law*

$$\eta_K := \sum_{(o, r): p_K(o, r) > 0} p_K(o, r) \delta_{(o, \pi_{or}^K)}. \quad (30)$$

It records the terminal payoff and the posterior about the realised action. Write $I_q(E)$ for the mutual information between the action and the complete visible output of E under q . When the output is a first-stage record Y , we also write this quantity as $I_q(A; Y)$.

Lemma 35 (Terminal laws and affine fibres). *Under Axioms (A1)–(A6), the following statements hold.*

- (a) *If q has full support and $\eta_K = \eta_L$, then $(q, K) \sim (q, L)$.*
- (b) *At every full-support q , the quotient of finite experiments by their marked terminal laws has an affine representation W_q .*

Proof. We give the finite Blackwell argument behind part (a). For a payoff o and posterior π , let

$$M_K(o, \pi) := \sum_{\substack{r: p_K(o, r) > 0 \\ \pi_{or}^K = \pi}} p_K(o, r).$$

Equality of marked laws gives $M_K(o, \pi) = M_L(o, \pi) =: M(o, \pi)$. For $p_K(o, r) > 0$, draw a record s of L according to

$$T(s | o, r) := \mathbf{1}_{\{p_L(o, s) > 0, \pi_{os}^L = \pi_{or}^K\}} \frac{p_L(o, s)}{M(o, \pi_{or}^K)}. \quad (31)$$

Choose an arbitrary probability row when $p_K(o, r) = 0$. This is a payoff-preserving record processor. Bayes' formula gives

$$q(a) K(o, r | a) = p_K(o, r) \pi_{or}^K(a).$$

For a reached target record s , with posterior $\pi = \pi_{os}^L$, it follows that

$$\sum_r K(o, r | a) T(s | o, r) = \frac{p_L(o, s)}{M(o, \pi)} \frac{\pi(a)}{q(a)} M(o, \pi) = L(o, s | a).$$

If s is unreached, full support makes both sides zero. Thus postprocessing K by T gives L . The symmetric construction gives a processor from L to K . Axiom (A4) and Lemma 34(a) give indifference. For a boundary prior, Lemma 34(b) first deletes the null actions.

For part (b), take the quotient of actual finite marked experiments by equality of (30). If E and G are two such experiments, their public mixture first draws a coin independent of the action, runs the selected experiment, and retains the coin in the record. Its marked law is

$$\eta_{tE \oplus (1-t)G} = t\eta_E + (1-t)\eta_G. \quad (32)$$

Axiom (A6) gives, for $0 < t < 1$ and any third experiment L ,

$$E \succeq_q L \iff tE \oplus (1-t)G \succeq_q tL \oplus (1-t)G. \quad (33)$$

For the reverse implication, if $E \not\succeq_q L$, completeness gives $L \succ_q E$; the strict part of Axiom (A6) makes the reverse public mixture strict. If record alphabets differ, put them in a common disjoint union before invoking the axiom; part (a) identifies the result with (32).

It remains only to check the Archimedean step. A closed segment between two experiments, and any fixed comparator, can be realised on one fixed union of their finite record alphabets. Its channel entries vary continuously with $t \in [0, 1]$. Axiom (A2) therefore makes both contour sets along the segment closed. Completeness says that they cover $[0, 1]$; when a target lies between the two endpoints, the sets contain opposite endpoints and connectedness makes them intersect. Thus every such target is indifferent to a point of the segment. The mixture-space theorem now supplies an affine representative W_q . This quotient is closed under every mixture and embedding used below. $\square$

B.2 The two common scales

Lemma 36 (Material utility and a common affine scale). *Under Axioms (A1)–(A7), there are outcomes o^+, o^- and a nonconstant function $u : O \rightarrow \mathbb{R}$ such that*

$$u(o^+) = 1, \quad u(o^-) = 0.$$

The payoff-lottery order is represented by $\sum_o \ell(o)u(o)$. The representatives in Lemma 35 may be normalised so that

$$W_q(\ell) = \sum_o \ell(o)u(o). \quad (34)$$

With this normalisation, adjoining a full-support independent dummy to a full-support prior preserves value, and for full-support q, q' ,

$$(q, E) \succeq (q', G) \iff W_q(E) \geq W_{q'}(G). \quad (35)$$

Proof. On a singleton action set, terminal-law sufficiency identifies an experiment with its payoff lottery. Public-mixture independence and continuity, proved as in Lemma 35, give an affine utility on $\Delta(O)$. Axiom (A7) supplies $o^+ \succ o^-$. Normalise their values to one and zero and let $u(o)$ be the value of the degenerate lottery at o . Affinity on the finite simplex gives expected utility.

An action report may ignore its input and draw an arbitrary target prior. Axiom (A5), applied in both directions, therefore makes this payoff lottery order independent of the prior and action alphabet. Normalise each W_q at the same two deterministic payoffs. The payoff-lottery embedding is affine and order-reflecting, so positive-affine uniqueness gives (34).

Dummy lifting between full-support fibres is likewise an affine order embedding. Positive-affine uniqueness initially permits

$$W_{q \otimes s}(E^\uparrow) = cW_q(E) + d, \quad c > 0.$$

The two payoff anchors have values zero and one on both sides; hence $c = 1$ and $d = 0$. To compare E at q with G at q' , lift each to the common product prior $q \otimes q'$, using the other prior as the dummy. Equation (29), the replacement part of Lemma 34, and the common product representative give (35). $\square$

We now use Proposition 19. Apply it to the constant-payoff lift

$$K_P(o, r \mid a) = \mathbf{1}_{\{o=o^-\}} P(r \mid a).$$

This lift commutes with block formation, record processing, action reporting, limits on each finite simplex, and compounding. Thus Axioms (A1) and (A8) give the pure weak order and its positive orientation; Axiom (A2) gives continuity; Axiom (A3) gives block coherence; Axioms (A4) and (A5) give the two data processing conditions, including every null-row completion; and Axiom (A6) gives branchwise continuation. These are precisely the hypotheses of Proposition 19. Its conclusion is that every within-channel and block-supported comparison between such lifts is ordered by $I_{q,P}(A; R)$.

Lemma 37 (The trace scale). *Under Axioms (A1)–(A8), there is one number $\lambda > 0$, independent of the prior and all finite alphabets, such that every full-support, nontrivial action fibre and every experiment E that pays o^- surely satisfy*

$$W_q(E) = \lambda I_q(E). \quad (36)$$

Proof. Fix a nontrivial full-support q . On the constant- o^- face, W_q is affine by Lemma 35. For any experiment E , the finite entropy identity

$$I_q(E) = H(q) - \sum_{(o,\pi) \in \text{supp } \eta_E} \eta_E(o, \pi) H(\pi). \quad (37)$$

shows that mutual information depends only on the marked terminal law and is affine under public mixtures. Proposition 19 says that W_q and I_q represent the same nonconstant order on the constant-low face. Positive-affine uniqueness therefore gives

$$W_q(K_P) = \lambda_q I_{q,P} + b_q, \quad \lambda_q > 0.$$

The uninformative experiment has mutual information zero and, by (34), value $u(o^-) = 0$. Thus $b_q = 0$.

The coefficient does not depend on q . Let q and q' be nontrivial full-support priors. Lift full revelation at q to $q \otimes q'$ by adding an independent dummy coordinate. Normalised value and mutual information are both unchanged. Since the common information value is $H(q) > 0$, cancellation gives $\lambda_{q \otimes q'} = \lambda_q$. Lifting full revelation from the other coordinate gives $\lambda_{q \otimes q'} = \lambda_{q'}$. Hence all such coefficients agree. Equivalently, fix the uniform prior on a two-point dummy alphabet and call its coefficient λ . Axiom (A8) fixes the strict orientation, so $\lambda > 0$.

Finally, any experiment that pays o^- surely is a pure record experiment after the constant payoff coordinate is deleted; restoring that coordinate changes neither its marked terminal law nor its mutual information. This proves (36). $\square$

B.3 The branch calculation

Let q be full support on a nontrivial A and let $P : A \rightarrow \Delta(Y)$ be a first-stage record channel. Put

$$m_y := \sum_a q(a) P(y | a).$$

For a deterministic payoff profile $g : Y \rightarrow O$, let $C(P, g)$ denote the compound that first runs P and then pays $g(y)$ with no further record. Write g_y^o for the profile that pays o at y and o^- elsewhere.

Lemma 38 (The reached-branch payoff increment). *For every $y \in Y$ and $o \in O$,*

$$W_q(C(P, g_y^o)) - W_q(C(P, o^-)) = m_y u(o). \quad (38)$$

Proof. If $m_y = 0$, changing the payoff on branch y does not change the marked terminal law, so both sides are zero. Suppose $m_y > 0$, and put

$$r_y(a) := \frac{q(a) P(y | a)}{m_y}, \quad S_y := \text{supp } r_y.$$

Let $\bar{r}_y$ be the full-support restriction of r_y to S_y . For a marked continuation E on S_y , extend E to A through a support projection, insert it at branch y , and put the uninformative payoff o^- at every other branch. Denote the resulting experiment by $J_y(E)$. If $a \notin S_y$, then $r_y(a) = 0$; since $q(a) > 0$ and $m_y > 0$, the definition of r_y gives $P(y | a) = 0$. Hence the continuation rows

chosen outside S_y are never used on branch y . Support restriction identifies their local versions as well, so the extension is independent of the chosen projection.

Support restriction and Axiom (A6), including its strict part, give

$$E \succeq_{\bar{r}_y} G \iff J_y(E) \succeq_q J_y(G).$$

For the reverse implication, failure of the local weak comparison gives a strict reverse comparison by completeness, and strict branchwise consistency gives a strict reverse compound comparison. Different continuation record alphabets are first put in a common disjoint union. Moreover, J_y commutes with public mixtures at the level of marked terminal laws. Thus affine uniqueness gives constants $c_y > 0, d_y$ such that

$$W_q(J_y(E)) = c_y W_{\bar{r}_y}(E) + d_y. \quad (39)$$

Suppose first that S_y has at least two elements. At payoff o^- , compare full revelation, E^{id} , with an uninformative record, E^0 . Lemma 37 gives the local difference

$$W_{\bar{r}_y}(E^{\text{id}}) - W_{\bar{r}_y}(E^0) = \lambda H(\bar{r}_y).$$

The finite mutual-information chain rule for insertion is

$$I_q(J_y(E)) = I_q(A; Y) + m_y I_{\bar{r}_y}(E). \quad (40)$$

Support extension does not change the last term. Both inserted experiments pay o^- surely, so Lemma 37 and (40) make their aggregate value difference $\lambda m_y H(\bar{r}_y)$. Equation (39) makes the same difference $c_y \lambda H(\bar{r}_y)$. Since $\lambda > 0$ and a nontrivial full-support distribution has positive entropy, $c_y = m_y$.

Locally, deterministic payoffs o and o^- have values $u(o)$ and zero by (34). Subtracting their two instances of (39), and identifying the resulting insertions with the two deterministic compounds in the statement, proves (38).

If S_y is a singleton, append an independent full-support two-point dummy action to the first-stage action. The reached posterior becomes the product of r_y and the dummy prior and hence has nontrivial support. The branch mass is still m_y , and Lemma 36 says that both compound values are unchanged. Apply the preceding argument in the product problem and remove the dummy. $\square$

Lemma 39 (Deterministic branch assembly). *For every deterministic payoff profile $g : Y \rightarrow O$,*

$$W_q(C(P, g)) = \lambda I_q(A; Y) + \sum_y m_y u(g(y)). \quad (41)$$

Proof. First observe that the increment from changing one branch is independent of the payoffs on the other branches. Indeed, for any two backgrounds g, h , the two crossed public half-mixtures

$$\frac{1}{2}C(P, g[y \mapsto o]) \oplus \frac{1}{2}C(P, h[y \mapsto o^-])$$

and

$$\frac{1}{2}C(P, g[y \mapsto o^-]) \oplus \frac{1}{2}C(P, h[y \mapsto o])$$

have the same marked terminal law. Away from y they contain the same two payoff-posterior atoms; at y they contain o and o^- with the same mass $m_y/2$. Affinity of W_q and cancellation of the two halves show that the two background-specific increments are equal. Taking h to be the all-low profile and applying Lemma 38, then changing the finitely many branches one at a time, gives

$$W_q(C(P, g)) - W_q(C(P, o^-)) = \sum_y m_y u(g(y)). \quad (42)$$

The all-low compound pays o^- surely and has exactly the information in the first-stage record. Lemma 37 therefore gives $W_q(C(P, o^-)) = \lambda I_q(A; Y)$. Substitute this into (42). $\square$

B.4 Proof of the theorem

Proof of Theorem 1. Assume first Axioms (A1)–(A8). Let q have full support on a nontrivial action set, and let $K : A \rightarrow \Delta(O \times R)$ be arbitrary. Put $Y = O \times R$ and take as first stage

$$P_K(y \mid a) := K(y \mid a), \quad y = (o, r).$$

After $y = (o, r)$, pay o for sure and reveal nothing further. Call the resulting compound K^{seq} . From K , the record processor sends (o, r) to the sequential record $((o, r), *)$. In the reverse direction it sends $((o, r), *)$ to r ; inconsistent payoff-record pairs are unreached and may be completed arbitrarily. Both processors preserve the actual payoff coordinate. Hence Axiom (A4) gives $(q, K) \sim (q, K^{\text{seq}})$.

Apply Lemma 39 with $g(o, r) = o$. The two identities needed are

$$I_q(A; Y) = I_{qK}(A; O, R)$$

and

$$\sum_{o,r} p_{qK}(o, r) u(o) = \mathbb{E}_{q,K}[u(O)],$$

respectively. Together with sequentialisation, they give

$$W_q(K) = \mathbb{E}_{q,K}[u(O)] + \lambda I_{qK}(A; O, R). \quad (43)$$

If A is a singleton, mutual information is zero and the marked terminal law is that of the induced payoff lottery. Equation (34) gives (43) directly. If q is on the boundary, put $S = \text{supp } q$ and write (q^S, K^S) for the restriction. Then

$$(q, K) \sim (q^S, K^S), \quad V(q, K) = V(q^S, K^S). \quad (44)$$

The first identity is Lemma 34(b); the second follows directly from the finite sums defining expected utility and mutual information. Thus no representative W_q is assigned to a boundary fibre.

Restrict two arbitrary alternatives to their supports, apply (35) and (43) there, and use (44) to transport the comparison back. This gives

$$(q, K) \succeq (p, L) \iff V(q, K) \geq V(p, L), \quad (45)$$

with the same u and λ . Axiom (A3), first by duplication and then by deletion of irrelevant blocks, turns (45) into the within-channel representation and the finite-block moreover clause. The construction made u nonconstant and $\lambda > 0$. This proves the forward implication.

Conversely, suppose that a nonconstant u and a number $\lambda > 0$ represent the family by V . Axiom (A1) follows from the order on $\mathbb{R}$. Expected utility and mutual information are jointly continuous in (K, q) on each finite simplex; for the latter, use the entropy formula and $0 \log 0 = 0$. Hence the set of triples (K, q, p) for which $V(q, K) \geq V(p, K)$ is closed, which is Axiom (A2). A block-supported prior has a constant block label, which changes neither term of V ; this proves Axiom (A3) and reduces every pair comparison below to the corresponding two values.

A payoff-preserving record processor leaves the payoff marginal unchanged and weakly lowers $I(A; O, R)$ by data processing. This proves Axiom (A4). Under an action report, equation (2) again leaves the payoff-record marginal unchanged, and the joint law forms the Markov chain

$$(O, R) - A - B.$$

Thus $I(B; O, R) \leq I(A; O, R)$, proving Axiom (A5). This argument uses the undivided joint-law equation and so also covers every completion on a null reported action.

For a compound channel, the two finite chain rules are

$$\begin{aligned}\mathbb{E}[u(O)] &= \sum_{y:m(y)>0} m(y)\mathbb{E}_{q_y, K^y}[u(O)], \\ I(A; Y, O, R) &= I(A; Y) + \sum_{y:m(y)>0} m(y)I_{q_y, K^y}(A; O, R).\end{aligned}$$

The first-stage term is common to two continuation profiles. Their aggregate value difference is therefore

$$\sum_{y:m(y)>0} m(y)[V(q_y, K^y) - V(q_y, L^y)].$$

Branchwise weak improvement makes this sum nonnegative, and a strict improvement on a reached branch makes it positive. This is precisely Axiom (A6).

Because u is nonconstant, two deterministic payoffs are strictly ranked; the trace term is zero on a singleton action set. This gives Axiom (A7). At a constant payoff, full revelation has value $u(o) + \lambda H(q)$ and an uninformative record has value $u(o)$. Full support on a nontrivial finite action set implies $H(q) > 0$; hence Axiom (A8). The same block calculation used above gives the moreover clause with the original witnesses u and λ . $\square$

C Auxiliary results

This appendix collects proofs of the uniqueness, matching, independence, and choice-implication results stated in the main text. None is used to prove Theorem 1; separating them keeps Appendices A and B focused on the representation argument.

C.1 Sign, uniqueness, and matching

Proof of Corollary 3. Part (i) is Theorem 1. For part (ii), reverse the primitive relation in every environment. The reversed family satisfies Axioms (A1)–(A3) and (A6); Axiom (A7) holds with its two witnessing outcomes interchanged. It also satisfies the forward versions of Axioms (A4), (A5), and (A8). Theorem 1 therefore represents the reversed family by $\mathbb{E}[\hat{u}(O)] + \hat{\lambda}I(A; O, R)$ with $\hat{\lambda} > 0$. Negating that index represents the original family, so it has the asserted form with $u = -\hat{u}$ and $\lambda = -\hat{\lambda} < 0$.

For part (iii), collapse the action alphabet of any pair (q, K) to a singleton. Axiom (A5⁰) makes the pair indifferent to the resulting one-action channel, whose output law is p_{qK} . Axiom (A4⁰) then erases its record without changing value. Thus every pair is indifferent to the one-action channel that delivers precisely its payoff marginal. Axiom (A3) makes these reductions coherent across blocks. On payoff lotteries, Axiom (A6) applied to an action-independent public coin gives mixture independence, Axiom (A2) gives continuity, and Axiom (A7) gives non-triviality. The mixture-space theorem therefore yields a nonconstant expected-utility index. Conversely, expected utility is unaffected by record or action processing and satisfies the three indifference clauses. The positive and negative formulae satisfy their respective clauses by data processing and the two chain rules. $\square$

Proof of Proposition 2. Choose $o_* \in O$ with $u(o_*) = \underline{u}$. First note that the range of V is $[\underline{u}, \infty)$. The lower bound is immediate. For $q \in \Delta(A)$ and $t \in [0, 1]$, let a constant- o_* record channel reveal the realised action with probability t and otherwise report $*$:

$$E_t(o_*, a' \mid a) = t\mathbf{1}_{a'=a}, \quad E_t(o_*, * \mid a) = 1 - t.$$

Then $V(q, E_t) = \underline{u} + \lambda tH(q)$. At the uniform lottery on n actions these values fill $[\underline{u}, \underline{u} + \lambda \log n]$; the intervals are unbounded.

For part (i), let W be a common-scale representation and define $\varphi(v) := W(q, K)$ for any pair with $V(q, K) = v$. This is well defined: equality of the two V -values gives indifference in their two-block environment by Theorem 1, and the common-scale property gives equality of the two W -values. A strict inequality between V -values similarly gives a strict inequality between W -values. Thus φ is strictly increasing and $W = \varphi \circ V$. The converse is immediate from Theorem 1.

For part (ii), the expected-utility and mutual-information chain rules give, for any two continuation profiles,

$$V(q, P \triangleright \{K^y\}) - V(q, P \triangleright \{L^y\}) = \sum_{y: m(y) > 0} m(y) [V(q_y, K^y) - V(q_y, L^y)].$$

Hence every $\alpha V + \beta$, with $\alpha > 0$, is common-scale and branch-additive.

Conversely, let the branch-additive common-scale representation be $W = \varphi \circ V$ and define

$$\psi(x) := \varphi(\underline{u} + x) - \varphi(\underline{u}), \quad x \geq 0.$$

Fix $n \geq 2$ and let q be uniform on n actions. For $x \in [0, \lambda \log n]$, choose a constant- o_* erasure channel whose value is $\underline{u} + x$. In the first continuation profile, use this channel after the first branch of a public coin with probabilities θ and $1 - \theta$, and use the uninformative constant- o_*

channel after the second. Let the comparison profile use the uninformative channel after both branches. The two compound values are $\underline{u} + \theta x$ and $\underline{u}$, so (6) gives

$$\psi(\theta x) = \theta \psi(x) \quad (0 \leq \theta \leq 1).$$

Taking $x = \lambda \log n$ shows that ψ is linear on $[0, \lambda \log n]$. These intervals are nested and unbounded, so $\psi(x) = \alpha x$ on $[0, \infty)$ for one $\alpha > 0$. Therefore $\varphi(v) = \alpha v + \beta$ on the range of V .

Finally, any second trace-tempered index $\tilde{V}$ is itself branch-additive. Applying part (ii) and then restricting to deterministic payoff outcomes and to constant-payoff record channels gives $\tilde{u} = \alpha u + \beta$ and $\tilde{\lambda} = \alpha \lambda$. $\square$

Proof of Corollary 4. The result follows from Theorem 1 and $I_{qK}(A; O, R) = H(q) - \int H(\pi) d\mu_{qK}(\pi)$. $\square$

C.2 Independence of the axioms

Proof of Proposition 5. Fix a nonconstant u and write $U(q, K) := \mathbb{E}_{q,K}[u(O)]$, $Z := (O, R)$, and $I := I(A; Z)$. All scores below are extended to block and compound channels by the same formula.

Failure of Axiom (A1). Use the Pareto relation on the two coordinates (U, I) :

$$(q, K) \succeq (p, L) \iff U(q, K) \geq U(p, L) \text{ and } I(q, K) \geq I(p, L).$$

It is transitive but incomplete whenever one alternative has more payoff and the other more information. Its graph is closed. Both data-processing axioms are coordinatewise monotonic, and both chain rules make branchwise weak and strict improvements aggregate. The two relevance axioms and block coherence hold.

Failure of Axiom (A2). Order alternatives lexicographically by (U, I) , with payoff first. This is a weak order and satisfies all structural, processing, continuation, and relevance requirements. It is not continuous even within one fixed channel. Let a three-action channel reveal the action fully and let only action 2 pay the high outcome. Put

$$p = (1/4, 1/2, 1/4), \quad q = (0.49, 0.50, 0.01), \quad q_n = (0.49 - \varepsilon_n, 0.50 + \varepsilon_n, 0.01),$$

where $\varepsilon_n \downarrow 0$. The lotteries p and q have the same expected utility but $H(q) < H(p)$, so $p \succ_K q$. Every q_n has strictly higher expected utility than p , so $q_n \succ_K p$. Thus the graph in Axiom (A2) is not closed.

Failure of Axiom (A3). Treat block labels as syntactic tags, with the outermost tag taking precedence. Put $w(a^0) = 1$, $w(a^i) = 0$ for $i \neq 0$, and $w(a) = 0$ on untagged actions. For $\varepsilon > 0$, use

$$V_\varepsilon(q, K) := U(q, K) + I(q, K) + \varepsilon \sum_a q(a)w(a).$$

Every record- or action-processing comparison places the original alternative in block 0, so the data-processing inequality receives the additional ε . If Δ_y is the base-score difference in continuation branch y , its block comparison is weak exactly when $\Delta_y + \varepsilon \geq 0$, while the aggregate block difference is

$$\sum_y m(y)\Delta_y + \varepsilon = \sum_y m(y)(\Delta_y + \varepsilon).$$

Thus Axiom (A6), including strictness, holds; the remaining axioms are immediate. Duplication coherence fails. Take constant payoff, K revealing a binary action, a degenerate q , and a nearby nondegenerate p with $0 < H(p) < \varepsilon$. Then $p \succ_K q$, but $q^0 \succ_{K \sqcup K} p^1$.

Failure of Axiom (A4). For $\varepsilon > 0$, use

$$V(q, K) := U(q, K) + I(A; Z) - \varepsilon H(Z | A). \tag{46}$$

This score is continuous and branch-additive, since both mutual information and conditional entropy obey the chain rule. Under action processing $A \rightarrow B$, the score difference between the original and processed alternatives is

$$[I(A; Z) - I(B; Z)] + \varepsilon[H(Z | B) - H(Z | A)] = (1 + \varepsilon)I(A; Z | B) \geq 0,$$

so (46) satisfies Axiom (A5). It also satisfies both relevance axioms. Record data processing fails: with constant payoff and a fair record independent of A , the original value is $-\varepsilon$; deleting that noise gives value zero.

Failure of Axiom (A5). Let $G(q) := 1 - \sum_a q(a)^2$ be Gini entropy and define its posterior reduction

$$J_G(q, K) := G(q) - \int G(\pi) d\mu_{qK}(\pi), \quad V := U + J_G.$$

Concavity of G gives record data processing, and the potential-difference form gives the exact branch chain rule. Full revelation is strictly valuable. Action processing can nevertheless increase J_G . Take $q = (1/8, 1/8, 3/4)$, a binary record with

$$\Pr(0 | a_1) = \Pr(0 | a_2) = 0, \quad \Pr(0 | a_3) = 1/4,$$

and merge a_1, a_2 into one reported action. Direct calculation gives $J_G = 9/416$ before the merge and $J_G = 3/104$ afterwards. Hence Axiom (A5) fails.

Failure of Axiom (A6). Use $V := U + \min\{I, 1\}$, with information measured in bits. It is continuous, respects both data-processing axioms, and has both strict orientations. Let q be uniform on four actions. A first-stage record reveals which of two pairs contains the action. In every branch, revealing the action within the pair strictly improves the continuation from zero to one bit. The two compounds have respectively one and two bits before capping and are therefore indifferent. The strict clause of Axiom (A6) fails.

Failure of Axiom (A7). Use $V := I$. All other axioms hold, but all payoff lotteries are indifferent.

Failure of Axiom (A8). Use $V := U$. All other axioms hold, but revealing the action at a fixed payoff is valueless. $\square$

C.3 Proofs of the choice implications

Results stated in the text only in summary form (Propositions 40 and 41, Corollary 42, and Remark 43) are restated here in full. Proofs appear in an order that front-loads shared machinery.

Proof of Corollary 11. Part (i) is the chain rule with an action-independent first stage: $I(A; Y) = 0$, and each reached branch reproduces the joint law of the corresponding constituent. Part (ii) follows from Proposition 12: deleting the coin coordinate is a deterministic payoff-preserving record processor carrying the public mixture to M_t . $\square$

Proof of Proposition 6. (i) Under either lottery, each visible pair (o_0, r_j) , $j = 1, 2$, has probability $\frac{1}{2}$: under q' because each a_j produces (o_0, r_j) for sure, under q'' because each of a_3, a_4 produces either pair with equal probability. Hence $p_{q'K} = p_{q''K}$. On $\text{supp}(q')$ the rows are degenerate and distinct, so the visible pair reveals the action and $I_{q'K} = H(q') = \log 2$; on $\text{supp}(q'')$ the rows are identical, so the visible pair is independent of the action and $I_{q''K} = 0$. The payoff terms agree because the payoff marginals do, giving the displayed gap. (ii) All rows of K° are equal, so $I_{q, K^\circ} = 0$ for every q ; since the payoff is constantly o_0 , the representation gives indifference. $\square$

Proof of Proposition 8. Expected utility is linear in q , while mutual information is concave in the input distribution of a fixed channel [18]. Continuity on the compact simplex gives existence. Using

$$I_{q, K}(A; Z) = \sum_a q(a) D_{\text{KL}}(K_Z(\cdot | a) \| m_{q, K}),$$

its directional derivative in the mass on action a equals the displayed divergence up to a term common to every action. The result is the Kuhn–Tucker condition for a concave programme. At constant u it reduces to the classical capacity condition. $\square$

Proof of Proposition 9. The entropy identity gives

$$V(q, K) = \lambda H(m_{q,K}) + \sum_a q(a) [\bar{u}(a) - \lambda H(P_a)].$$

If q_0 and q_1 are optimal with $m_{q_0,K} \neq m_{q_1,K}$, strict concavity of Shannon entropy and linearity of $q \mapsto m_{q,K}$ and of the remaining sum give $V(tq_0 + (1-t)q_1, K) > C^*(K)$ for $t \in (0, 1)$, a contradiction. Hence every optimum induces the same m^* . By Proposition 8, an optimum q^* satisfies the displayed equalities at some constant c ; averaging them under q^* gives $c = V(q^*, K) = C^*(K)$, so $\text{supp}(q^*) \subseteq \mathcal{A}^*$. Conversely, if $m_{q,K} = m^*$ and $\text{supp}(q) \subseteq \mathcal{A}^*$, then

$$V(q, K) = \sum_a q(a) [\bar{u}(a) + \lambda D_{\text{KL}}(P_a \| m^*)] = C^*(K).$$

Affine independence makes the representation $m^* = \sum_a q(a)P_a$ with $\text{supp}(q) \subseteq \mathcal{A}^*$ unique. Identical rows share $\bar{u}(a)$ and $H(P_a)$, so both $m_{q,K}$ and the displayed expression for V are unchanged by redistribution within a class; collapsing the classes and lifting aggregate masses proves the last claim. $\square$

Proof of Proposition 7. Throughout, m^* denotes the common optimal marginal of Proposition 9.

(i) Suppose $m^*(z) = 0$ for some $z \in Z_K$ and choose a with $K_Z(z | a) > 0$. Then $D_{\text{KL}}(K_Z(\cdot | a) \| m^*) = +\infty$, which is incompatible with the finite on-support equality if some optimum uses a and with the off-support inequality of Proposition 8 if none does. Under the private-consequence condition, an optimum with $q^*(a) = 0$ would force $m^*(z_a) = 0$.

(ii) For $q = \delta_a$ the induced marginal is $K_Z(\cdot | a)$ and the on-support constant is $\bar{u}(a)$; the displayed system is the off-support condition of Proposition 8, necessary and sufficient by concavity. Under strict inequalities, $\mathcal{A}^* = \{a\}$ in Proposition 9, so every optimum is δ_a . $\square$

Proposition 40 (Optimal mixing). *Let $A = \{a, b\}$ with $K_a := K_Z(\cdot | a) \neq K_b := K_Z(\cdot | b)$ and $\Delta := \bar{u}(a) - \bar{u}(b) \geq 0$. For $x \in [0, 1]$ let q_x put mass x on b , so that $m_{q_x,K} = (1-x)K_a + xK_b =: m_x$, and for $x \in (0, 1)$ define*

$$g(x) := D_{\text{KL}}(K_b \| m_x) - D_{\text{KL}}(K_a \| m_x).$$

- (i) *g is continuous and strictly decreasing, with $g(0^+) = D_{\text{KL}}(K_b \| K_a)$ and $g(1^-) = -D_{\text{KL}}(K_a \| K_b)$ as extended-real limits. If $\Delta < \lambda D_{\text{KL}}(K_b \| K_a)$, the optimum is unique and interior, $q^* = q_{x^*}$ with x^* the unique solution of $\Delta = \lambda g(x)$.*
- (ii) *The optimal mass $x^*(\Delta, \lambda)$ on the inferior action is nonincreasing in Δ and nondecreasing in λ , strictly so on the mixing region; at $\Delta = 0$ it is the capacity-achieving input of the two-row channel, independently of λ .*

Proof. Write $f(x) := V(q_x, K) = \bar{u}(a) - x\Delta + \lambda I(x)$ with $I(x) = H(m_x) - (1-x)H(K_a) - xH(K_b)$. Since $K_a \neq K_b$, $x \mapsto m_x$ is affine and injective, so strict concavity of H [18] makes f strictly concave on $[0, 1]$, and a unique maximiser exists. On $(0, 1)$ every $z \in \text{supp } K_a \cup \text{supp } K_b$ has $m_x(z) > 0$ and, using $\sum_z (K_b(z) - K_a(z)) = 0$,

$$I'(x) = H(K_a) - H(K_b) - \sum_z (K_b(z) - K_a(z)) \log m_x(z) = g(x),$$

so $f'(x) = -\Delta + \lambda g(x)$, with g continuous and strictly decreasing as the derivative of a strictly concave function. As $x \downarrow 0$, $m_x(z) \rightarrow K_a(z)$; if $\text{supp } K_b \subseteq \text{supp } K_a$ every summand converges and

$g(0^+) = D_{\text{KL}}(K_b \| K_a)$, while if some $\bar{z}$ has $K_b(\bar{z}) > 0 = K_a(\bar{z})$ the summand $-K_b(\bar{z}) \log(xK_b(\bar{z}))$ diverges to $+\infty$, matching the extended-sense divergence. Exchanging the roles of a and b gives the limit at 1. By concavity and the mean value theorem, δ_a is optimal if and only if $\lim_{x \downarrow 0} f'(x) \leq 0$, i.e. $\Delta \geq \lambda D_{\text{KL}}(K_b \| K_a)$; this is the threshold stated in the text, and Proposition 7(ii) with two actions. If $\Delta < \lambda D_{\text{KL}}(K_b \| K_a)$, the maximiser is not $x = 0$ by the threshold and not $x = 1$ because $f'(1^-) = -\Delta - \lambda D_{\text{KL}}(K_a \| K_b) < 0$; it is therefore the unique root of $\Delta = \lambda g(x)$, which exists because g decreases strictly between its limits. This proves (i). The comparative statics of (ii) follow by inverting the strictly decreasing g at Δ/λ : raising Δ raises the target, raising λ lowers it; at $\Delta = 0$ the condition $g(x) = 0$ is the capacity condition of Proposition 8 at constant payoff. $\square$

Proposition 41 (Trace demand). *Fix K and write $e(q) := \mathbb{E}_{q,K}[u(O)]$, $i(q) := I_{qK}(A; O, R)$,*

$$V^*(\lambda) := \max_{q \in \Delta(A)} \{e(q) + \lambda i(q)\}, \quad Q(\lambda) := \arg \max_{q \in \Delta(A)} \{e(q) + \lambda i(q)\}.$$

(i) V^* is convex on $(0, \infty)$, i is affine on the convex compact set $Q(\lambda)$, and

$$(V^*)'_-(\lambda) = \min_{q \in Q(\lambda)} i(q), \quad (V^*)'_+(\lambda) = \max_{q \in Q(\lambda)} i(q);$$

at every point of differentiability, $(V^*)'(\lambda) = i(q)$ for every $q \in Q(\lambda)$.

(ii) Let $0 < \lambda_1 < \lambda_2$ and $q_j \in Q(\lambda_j)$, writing $e_j := e(q_j)$, $i_j := i(q_j)$. Then $i_1 \leq i_2$, $e_1 \geq e_2$, and

$$\lambda_1(i_2 - i_1) \leq e_1 - e_2 \leq \lambda_2(i_2 - i_1).$$

Hence either $(e_1, i_1) = (e_2, i_2)$, or $i_1 < i_2$ and $e_1 > e_2$ and

$$\lambda_1 \leq \frac{e_1 - e_2}{i_2 - i_1} \leq \lambda_2.$$

Proof of Proposition 41. (i) V^* is a pointwise maximum of functions affine in λ , hence convex and, since $0 \leq i \leq \log |A|$, finite. On the convex compact set $Q(\lambda)$ the objective is constant and e is linear, so $i = (V^* - e)/\lambda$ is affine there. For $q \in Q(\lambda)$ and $\mu > 0$, $V^*(\mu) \geq V^*(\lambda) + (\mu - \lambda)i(q)$, so each such $i(q)$ is a subgradient. For the right derivative, choose $q_h \in Q(\lambda + h)$; then

$$\max_{q \in Q(\lambda)} i(q) \leq \frac{V^*(\lambda + h) - V^*(\lambda)}{h} \leq i(q_h),$$

the first inequality by the subgradient bound, the second by optimality of $Q(\lambda)$ at λ . By compactness and continuity, accumulation points of $(q_h)_{h \downarrow 0}$ lie in $Q(\lambda)$, so the right-hand side accumulates in $i(Q(\lambda))$ and the right derivative equals the maximum. The left derivative is symmetric, and a convex function on an interval is differentiable off a countable set.

(ii) Optimality at the two weights gives $e_1 + \lambda_1 i_1 \geq e_2 + \lambda_1 i_2$ and $e_2 + \lambda_2 i_2 \geq e_1 + \lambda_2 i_1$. Adding yields $(\lambda_2 - \lambda_1)(i_2 - i_1) \geq 0$, hence $i_2 \geq i_1$; the two inequalities rearrange to the displayed bracket, which gives $e_1 \geq e_2$ and, when the pairs differ, the final bound. $\square$

Corollary 42 (Extreme trace weights). *Let $e^0 := \max_q e(q)$, $i^0 := \max\{i(q) : e(q) = e^0\}$, $C := \max_q i(q)$, and $e^C := \max\{e(q) : i(q) = C\}$. If $\lambda_n \downarrow 0$ and $q_n \in Q(\lambda_n)$, every accumulation point of (q_n) maximises i among the maximisers of e ; if $\lambda_n \uparrow \infty$, every accumulation point maximises e among the maximisers of i . Moreover*

$$V^*(\lambda) = e^0 + \lambda i^0 + o(\lambda) \quad \text{as } \lambda \downarrow 0, \quad V^*(\lambda) = \lambda C + e^C + o(1) \quad \text{as } \lambda \uparrow \infty.$$

Proof of Corollary 42. For $\lambda_n \downarrow 0$, comparing q_n with an e -maximiser attaining i^0 gives

$$0 \leq e^0 - e(q_n) \leq \lambda_n(i(q_n) - i^0).$$

Boundedness of i forces $e(q_n) \rightarrow e^0$, and the same inequality gives $i(q_n) \geq i^0$; continuity delivers the accumulation claim. Dividing by λ_n ,

$$\frac{V^*(\lambda_n) - e^0}{\lambda_n} = i(q_n) - \frac{e^0 - e(q_n)}{\lambda_n},$$

where the last fraction lies in $[0, i(q_n) - i^0]$ and vanishes, proving the first expansion. For $\lambda_n \uparrow \infty$, comparing q_n with an i -maximiser attaining e^C gives $0 \leq \lambda_n(C - i(q_n)) \leq e(q_n) - e^C$, and the symmetric argument completes the proof. $\square$

Proof of Proposition 12. The processor leaves the payoff coordinate unchanged, so the expected-utility terms coincide and the difference is $\lambda[I_{qK}(A; O, R) - I_{qL}(A; O, R')]$. Write $Z := (O, R)$ and $Z' := (O, R')$. By construction $A \rightarrow Z \rightarrow Z'$ is a Markov chain, so expanding $I(A; Z, Z')$ in both orders gives

$$I(A; Z) - I(A; Z') = I(A; Z | Z') = I(A; R | O, R'),$$

the last equality because O is part of Z' . Conditional mutual information is nonnegative and vanishes exactly under the conditional independence in (b) [18], proving the identity and (a) $\Leftrightarrow$ (b). For (b) $\Rightarrow$ (c): the Markov property gives $\Pr(A = \cdot | Z, Z') = \pi_Z^{qK}$ almost surely, while (b) gives $\Pr(A = \cdot | Z, Z') = \pi_{Z'}^{qL}$; the two random posteriors coincide almost surely, so their laws are equal. For (c) $\Rightarrow$ (a): by the identity displayed after Corollary 4, the information term depends on (q, K) only through μ_{qK} , and the payoff terms already agree. $\square$

Proof of Proposition 18. (i) If $L = KT$, Proposition 12 gives $I_{qK} \geq I_{qL}$ at every q . Because the payoff kernels coincide, $V(q, K) - V(q, L) = \lambda[I_{qK} - I_{qL}]$ with $\lambda > 0$, so uniform value dominance and $K \succeq_c L$ are the same statement.

(ii) Let $A = \{0, 1\}$, fix $o^* \in O$, and let both channels pay o^* surely. Let K carry the binary erasure record with erasure probability $\varepsilon = \frac{1}{2}$ and L the binary symmetric record with crossover $p = \frac{1}{5}$:

$$K(o^*, a | a) = 1 - \varepsilon, \quad K(o^*, e | a) = \varepsilon, \quad L(o^*, a | a) = 1 - p, \quad L(o^*, 1 - a | a) = p.$$

The payoff kernels agree row by row, so it suffices to compare informations, and the comparison is invariant to the base; compute in bits with h the binary entropy. For $q = (1 - t, t)$, grouping records gives

$$I_{qK} = (1 - \varepsilon)h(t), \quad I_{qL} = h(p * t) - h(p), \quad p * t := p(1 - t) + (1 - p)t.$$

By Mrs Gerber's Lemma [49], $v \mapsto h(p * h^{-1}(v))$ is convex on $[0, 1]$, with h^{-1} valued in $[0, \frac{1}{2}]$; since $h(p * t)$ is invariant under $t \mapsto 1 - t$, take $t \leq \frac{1}{2}$ and $v = h(t)$. The chord between $v = 0$ and $v = 1$ gives

$$h(p * t) \leq (1 - v)h(p) + v, \quad \text{hence} \quad I_{qL} \leq (1 - h(p))h(t) \leq (1 - \varepsilon)h(t) = I_{qK},$$

the last step because $\varepsilon = \frac{1}{2} \leq h(\frac{1}{5})$. Thus $K \succeq_c L$. Now suppose $L = KT$ for a processor T , and write $t_r := T(0 | o^*, r)$ for $r \in \{0, 1, e\}$. Matching the probability of the processed record 0 in the two rows,

$$(1 - \varepsilon)t_0 + \varepsilon t_e = 1 - p, \quad (1 - \varepsilon)t_1 + \varepsilon t_e = p,$$

and subtracting gives $t_0 - t_1 = (1 - 2p)/(1 - \varepsilon) = \frac{6}{5} > 1$, impossible. Compositions of record processors are record processors, so no chain of garblings produces L from K either. $\square$

Proof of Proposition 14. (i) The material part of the compound is $(1 - \varepsilon)\mathbb{E}_{q,K}[u(O)] + \varepsilon w(\beta)$. For the trace part, the coin is action-independent, so both branch posteriors equal q and the chain rule for the compound gives an information term of $(1 - \varepsilon)I_{qK}(A; O, R)$, the sweetener branch contributing zero. (ii) By the moreover clause of Theorem 1, the indifference holds if and only if $(1 - \varepsilon)V(q, K) + \varepsilon w(\beta) = (1 - \varepsilon)V(p, L) + \varepsilon w(\beta')$; rearrange, and substitute the representation when the material parts cancel. (iii) $w(\beta) - w(\beta')$ is continuous on $[0, 1]^2$ with range $[-\Delta u, \Delta u]$; apply the intermediate value theorem to the equation in (ii). $\square$

Proof of Proposition 15. (i) Inclusion is immediate. Conversely, fix (t, w) in the strip, choose n with $\log n \geq t$, and on $A = \{1, \dots, n\}$ use continuity of entropy along a segment from a point mass to the uniform lottery to find q_t with $H(q_t) = t$. Let $K(o^+, a | a) = \beta$ and $K(o^-, a | a) = 1 - \beta$ with β chosen so that $w(\beta) = w$. The payoff lottery is action-independent with mean w , and the record reveals the action, so $\phi(q_t, K) = (t, w)$. The description of the indifference classes then reads the representation through ϕ , using the moreover clause of Theorem 1. (ii) The identity rearranges $E_{qK} + \lambda I_{qK} = E_{pL} + \lambda I_{pL}$. For uniqueness, let $\lambda' \neq \lambda$, put $t := \frac{1}{2} \min\{1, (\bar{u} - \underline{u})/\lambda\} > 0$, and use (i) to realise pairs with ϕ -coordinates $(t, \bar{u} - \lambda t)$ and $(0, \bar{u})$; they are indifferent, while $E + \lambda' I$ differs across them by $(\lambda - \lambda')t \neq 0$. $\square$

Remark 43 (Eliciting λ). Normalise $u(o^+) = 1$ and $u(o^-) = 0$. All comparisons below are block comparisons, hence within the primitive domain, and no information quantity is estimated: the information gap is an accounting identity of the design. *Step 1.* For each $o \in O$, elicit β_o with $(\delta_*, K_o) \sim (\delta_*, B_{\beta_o})$, where K_o is the degenerate one-action channel of Axiom (A7); one-action channels carry zero information, so $u(o) = \beta_o$. *Step 2.* Fix $o \in O$, $n \geq 2$, the uniform q on $\{1, \dots, n\}$, a dilution $\varepsilon \in (0, 1)$, and the constant-payoff channels K_o^{id} and K_o^0 of Axiom (A8). Elicit β with

$$(q, D_{\varepsilon, \beta} K_o^0) \sim (q, D_{\varepsilon, 0} K_o^{\text{id}}); \quad \text{then} \quad \lambda = \frac{\varepsilon}{1 - \varepsilon} \cdot \frac{\beta}{\log n},$$

by Proposition 14(i) applied to $V(q, K_o^{\text{id}}) = u(o) + \lambda \log n$ and $V(q, K_o^0) = u(o)$. Such a β exists if and only if $\lambda \log n \leq \varepsilon/(1 - \varepsilon)$; a subject still strictly preferring the right block at $\beta = 1$ reveals $\lambda > \varepsilon/((1 - \varepsilon) \log n)$, and ε is raised. *Controls.* Orientation: $(q, K_o^{\text{id}}) \succ (q, K_o^0)$ is Axiom (A8) itself. Garbling: under the total record garbling of K_o^{id} , both blocks carry zero information and indifference must obtain; a surviving strict preference indicates that an uncontrolled feature of the display still traces the action. Transport: repeating Step 2 at another n , prior, or payoff level must return the same λ , by Proposition 15(ii); an estimate that declines in $\log n$ is the signature of the capped-information criterion.